

Liberia National ECD Curriculum: Teacher Planner

3- to 6-Year-Olds

MY ENVIRONMENT: May and June



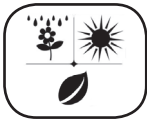
Ministry of Education / Bureau of Early Childhood Education

2017

INTRODUCTION TO MY ENVIRONMENT

The content area “**My Environment**” is planned for two months approximately – **May and June** and includes the following topics, each of them might last one – two weeks depending on the interest of children:

My Environment

I.		Seasons and Weather 1 week	1
II.		Plants 2 weeks	15
III.		Animals: Farm Animals 2 weeks	35
IV.		Animals: Small Animals 2 weeks	59
V.		Animals: Tiny Creatures 1 week	78



I: SEASONS AND WEATHER

BACKGROUND INFORMATION

Young children tend to find the concept of seasons rather difficult because it is too abstract, but it can be easier to understand since there are two seasons in Liberia – the wet (rainy), which lasts from May to October, and the dry season from November to April.

Within this theme the children can learn that the weather changes according to the seasons: some seasons are rainy, some are dry; wind and clouds in the sky bring rain, sometimes thunder and lightning; the amount of rain that falls and the temperature can be measure and we can keep records of the weather.

Teachers can encourage children to observe the weather conditions – how the sky changes in color and appearance during different weather conditions. Children can experience the wind and how it changes in direction and speed. In rainy weather children can look at the puddles and what happens to them after the rain over a few days. It is interesting to watch the clouds change shape as they are blown by the wind. Sometimes they look like faces or animals.

The following information about the weather might be useful for teachers:

- **Clouds:** different kinds of clouds have different names, e.g. the dark rain clouds are called nimbus clouds.
- **Thunderstorms** are caused by giant electric sparks between the clouds and the earth (fork lightning), or inside the clouds (sheet lightning). The electric spark creates great eat which makes the air expand very quickly. This makes a loud echoing noise-thunder.
- **Rainbows** occur when sunlight falls on drops of water in rain, fog or spray because the droplets split light into seven colors that it is made up of – red, orange, yellow, green, blue, indigo and violet. Sometimes you can see them in a bubble.



WEEKLY PLANNING OVERVIEW – May, Week 1

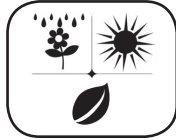
I: Seasons and Weather

Time	Type of Activity	Monday	Tuesday	Wednesday	Thursday	Friday
9.00 – 9.15	Morning Meeting	• Welcome children • Greetings • What is the topic for the week? • Introducing Weather p. 2	• Welcome children • Greetings • What is the topic for the week? • Introducing Weather p. 2	• Welcome children • Greetings • Making Rain p. 3	• Welcome children • Greetings • Making Rain p. 3	• Welcome children • Greetings • Shadows p. 4
9.15 – 11.00	Work Time – Learning Centers: Teacher organizes materials for children to work in centers of the child's choice including: Art Center, Block Center, Drama Center, Literacy Center, Manipulative Center while one or two Small Groups are working on teacher-initiated activities.					
	Small Groups Week 1	Water Thermometer p. 8	Weather Reports p. 10	Water Thermometer p. 8	Weather Reports p. 9	Rain Gauge p. 11
11.00 – 11.15	Clean-Up					
11.15 – 12.00	Outdoor Time	Clouds p. 13	Rain Gauge p. 11	Season Walks p. 12	Free Play	Season Walks p. 12
12.00 – 12.45	Wash-Up, Lunch					
12.45 – 14.15	Rest & Quiet Time	Children who can't sleep may "read" books.				
14.15 – 14.30	Group Story Time	Social games, story reading, finger plays, singing, dramatization of the books, reading the children's books related to the topic.				
14.30 – 15.00	Closing Meeting	Review of the day, sharing impressions of the day, discussion of the activities children have been involved in – which new things they learned and planning the next day.				

Activities with Parents throughout the day	• Parents can volunteer in Small Group activities • Parents can join in songs, games, books, storytelling • Parents can learn an activity and practice at home
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A. IDEAS FOR THE MORNING MEETING



Introducing the Weather

Objectives:

- To introduce the theme and discuss weather characteristics of each season.

Materials needed:

- Season posters and pictures
- Or markers and paper to make season pictures

Preparation:

1. Locate a vehicle to be brought to the Center or a vehicle that the children can be brought to.

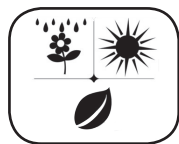
Instructions:

All Ages

1. Encourage the children to talk about the clothing they are wearing to school and why. Talk about the weather and ask if it is the same as the day before. Is it always the same? Introduce the concept of the particular season, e.g. the rainy time, the dry time.
2. Talk generally about how the season affects people's lives, how we try to keep cool or dry, why the rain is important.

Use the following questions to get the children talking.

- *How do we dress when it is raining / hot? Why?*
- *How do we protect ourselves from the sun /rain?*
- *What do you think it is made of? Why?*
- *What would happen if the raincoat was made of wool?*
- *What can we wear when it is sunny? Why?*



Making Rain

Objectives:

- To learn more about rain.

Materials needed:

- Ice (if available)
- Coal pot and fire coal/stove
- Rain gauge
- Kettle or pot of boiling water
- Sponges and water

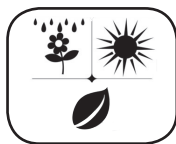
Instructions:

Beginner:

1. On a rainy day take the children out so that they can observe the falling rain. Let them feel it on their hands and faces. Ask them to close their eyes and listen to the sounds the rain makes when it hits the various objects. Talk about how it looks, smells and tastes.

Intermediate/Advanced

1. On a rainy day take the children out so that they can observe the falling rain. Let them feel it on their hands and faces. Ask them to close their eyes and listen to the sounds the rain makes when it hits the various objects. Talk about how it looks, smells and tastes.
2. Fill a pot with ice and hold it over the boiling kettle or a pot of boiling water. Tell the children to watch carefully. As the hot steam hits the cold saucepan, the vapor will condense into large droplets which fall like rain.
3. Talk about how the rain that we see is made in the same way as this. The water from the earth's surface evaporates into the air as water vapor, especially when it is hot (things get dry). Clouds form in cold air. When the clouds become too full of water, it falls back to the earth as rain.
4. Let the children play with the sponges and water, letting the sponges get full of water and then squeezing them to release the water.
5. When it has stopped raining, go outside and check the rain gauge (see small group activity pg. 11 for making rain gauge). Talk about how much it has rained and how to measure it.
6. Think of an indoor activity so that children who have health issues can enjoy a similar experience.



Shadows

Objective:

- To talk about the sun and how shadows are made.
- To observe and discuss changes that occur during a day that shows the passing of time.

Materials needed:

- Piece of chalk or ball of string

Instructions:

Beginner:

1. Take children outside in the morning when there is strong sunlight. Explain to them that the sun rises in the east. Ask them to show you where they think the east might be. As it rises, it throws long shadows to the west (where do they think west is?). Discuss East, West, North, South.
2. Make a game of jumping on your own shadow. Draw or mark around each child's feet. Then draw or mark out the outline of the child's shadow.
3. Tell the children to try and jump on their shadow. The children will soon find that it can't be done, but they will enjoy trying. They can try jumping on someone else's shadow. Can they make their shadows dance? Let them experiment with shadows. What different shapes can they make with their bodies?

Intermediate/Advanced:

1. Take children outside in the morning when there is strong sunlight. Explain to them that the sun rises in the east. Ask them to show you where they think the east might be. As it rises, it throws long shadows to the west (where do they think west is?). Discuss East, West, North, South.
2. Draw or mark around each child's feet. Then draw or mark out the outline of the child's shadow. Come back at different times of the day (midday and mid-afternoon) and talk about how the shadow has changed. Discuss why children think the shadow changes and why there is hardly any shadow at midday (in summer).
3. Make a game of jumping on your own shadow. The children will soon find that it can't be done, but they will enjoy trying. They can try jumping on someone else's shadow. Can they make their shadows dance? Let them experiment with shadows. What different shapes can they make with their bodies?



4. Use the following questions to get the children talking
- *What shape do you see?*
 - *How is it different?*
 - *Why do you think it has changed?*
 - *How can you make your shadow longer / rounder?*
 - *What would happen to your shadow if you stood under the tree? Why?*
 - *What would happen on a cloudy day?*

B. IDEAS FOR THE SMALL GROUP ACTIVITIES



Newspaper Hats

Objectives:

- To provide protection from the hot sun.
- To encourage creativity.

Materials needed:

- Newspapers
- Paint and paintbrushes
- Stapler or tape, scissors

Instructions:

Beginner:

1. Explain to the children that they are going to make hats to protect them from the sun. Ask each child to choose which hat to make. Give a full sheet of newspaper for the triangular hat and a half a sheet of newspaper for the round hat.
2. Encourage Children to decorate their paper with the paint. You will need to help children with making their hats. Children can wear their hats when they go outside to play.

Intermediate/Advanced:

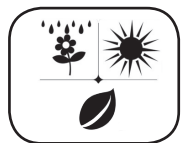
1. Explain to the children that they are going to make hats to protect them from the sun. Ask each child to choose which hat to make. Give a full sheet of newspaper for the triangular hat and a half a sheet of newspaper for the round hat.
2. Children decorate their paper with the paint. You will need to help children with making their hats. Children can wear their hats when they go outside to play.
3. Older children can be shown the following steps and they can cut and staple their own hats.

**Triangular hat:**

1. Fold the piece of newspaper in half and make a crease.
2. Fold in half again and crease Undo the last fold.
3. Fold the corners of the closed section towards the middle crease to form a triangle leaving a strip of about 8 cm at the bottom.
4. Fold each strip back separately, and stick it with tape or paste, if necessary. Open to make hat.

Round hat:

1. Fold the piece of newspaper in half and make a crease-
2. Fold in half again and crease.
3. Undo the last fold.
4. Cut two-thirds of the way along the crease line of the doubled sheet towards the fold.
Then cut down at right angles to the edge.
5. Undo the first fold.
6. Fold at the front to make a brim.
7. Staple x to x and z to z.

**Water Patterns****Objectives:**

- To use natural materials (water and earth) to make patterns.

Materials needed:

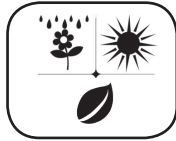
- Tins with one, two or three small holes in the lids or plastic bottles with a small hole in the bottle top
- Water
- Flat, dirt space outdoors OR indoor sand/dirt box

Instructions:**Beginner:**

1. Provide a cleared dirt space, and plastic bottle with water in it. Show the children how to drip water into the dirt/sand. Let the children have fun creating patterns

Intermediate/Advanced:

1. Ask each child to pick an area of earth and tell them to move all stones, twigs and leaves to the side, creating a smooth space. Tell the child to draw a circle around their smooth space (this can be done indoors if an indoor sand/dirt area is available).
2. Show the children how they can make patterns in their circles by dripping water from a bottle or tin onto the smooth earth.
3. They can also experiment to see what kinds of patterns they can make using a tin (or bottle) with more than one hole in it.



Water Thermometer

Objectives:

- To discover the effects of temperature changes and to measure air temperature.

Materials needed:

- Glass or clear plastic bottle, piece of rubber
- Transparent tube (a clear plastic straw will do)
- Cardboard or paper, pen, ruler, scissors, glue
- Food coloring (e.g. vegetable dye)
- Variety of wooden, metal and plastic things, white and black plastic and cloth things,
- Water in a glass jar

**Preparation:**

1. Half fill the bottle with cold water and color it with the food coloring.
2. Tie the piece of rubber over the neck of the bottle. Make a hole in the rubber and push the tube through it so that it fits tightly and almost reaches the bottom of the bottle.
3. Mark a piece of white card or paper like the markings on a thermometer. Stick the card on one side of the bottle with the markings facing inward.

Instructions:**All Ages:**

1. Do this activity on a very hot day. Put out the variety of things (plastic, wooden, metal, and black and white). Encourage the children to feel the temperature of the things and talk about which feel the same temperature, warm or cool, warmer, warmest / cooler, coolest. Then ask them to predict which things will get hotter than others when put in the sun. Ask them to put the things in the sun, including the jar of water.
2. After a while (perhaps the children can look around outside for things that are hot and cool while they are waiting and compare how they feel) the children can feel the things they put in the sun to discover if their predictions were correct. Ask them why some things get hotter than others (some soak up heat, like metal and black things}. They can group the various things or arrange them from coolest to hottest
3. Suggest that they feel the temperature of the water in the jar. Has it got warmer? Then show them the water thermometer and explain that it can be used to measure changes in the air temperature. (As it gets hotter, the water expands and moves up the tube, and as the water cools, it contracts and the level of the water in the tub falls. It can be placed in the sun and in the refrigerator (if available) to show the changes.



Weather Reports

Objectives:

- To encourage children to observe weather changes and create their own ways of recording weather conditions.

Materials needed:

- Weather paper dolls
- Pencil
- Crayons
- Water thermometer and rain gauge paper
- Ruler
- Scissors

Instructions:

Beginner: • Note: Not for younger children.

Intermediate/ Advanced

1. Encourage older children to make their own weather reports using whatever they like. For example, they can draw, paint and/or cut and stick; they can draw the sun, clouds or rain, and they can dress a 'weather' paper doll in suitable clothes and stick it onto their weather report, or draw their own person.
2. Also encourage them to record the temperature or amount of rain in some way. They could use measuring strips or draw the water thermometer showing the height of the water in the tube. Keep the water thermometer on a window sill where it gets the morning sun.
3. In group time the children who have made weather reports can show these to the group and talk about them. They can also compare their earlier and later measurements. Has the day got hotter or cooler? Why?



Rain Gauge

Objectives:

- To measure and record the amount of rain that has fallen.

Materials needed:

- Clear plastic round bottle with smooth sides
- Sharp knife or scissors
- Paper
- Ruler
- Pens, markers, pencils, crayons

Preparation:

1. Make the Rain Gauge. Remove the lid of the bottle and cut off the top third of the bottle. Place the top part upside down in the bottle to act as a funnel.
2. Mark off measuring lines on a strip of paper with ruler and pen. Stick this strip to the bottle so that children will be able to see how much rain water is inside the bottle.
3. Cut a number of strips of paper and mark off measuring lines in the same way as those on the gauge to use for recording purposes.

Instructions:

Beginner:

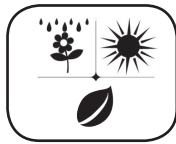
1. Place rain gauge in an open area outside on a flat surface.
2. Check the rain gauge after each rainfall and record on a chart on the wall during the rainy season. Remember to empty the rain gauge after each reading.

Intermediate/ Advanced

1. Place the bottle in an open area outside on a flat surface. Remember to empty the rain gauge after each reading. Do this activity in the rainy season.
2. Help the children record the rainfall in different ways. They can count how many long lines the water 'covers' on the measuring guide and then a child can color in one of the separate strips up to the same mark (it can be checked by holding it against the bottle). Display the strips in some way so that they can be compared,

3. A more concrete way to record rainfall is to mark the height on the bottle, either by sticking a small strip of paper at the height of the water or using elastic bands to mark the height each time it rains.
4. Encourage the children to think of other ways of recording the rainfall, e.g. they could draw rain gauges showing the height of the water. Older children might try to use a ruler to measure properly and to write numerals.

C. IDEAS FOR OUTDOOR GAMES



Season Walks

Objectives:

- To observe plants, animals, farming activities and changes in the surroundings according to the season.

Materials needed: None

Preparation:

1. Plan the walk to places where children will see changes in the trees, crops, flowers, and in the farm animals or tiny creatures that live in the surroundings.
2. If there are special farming activities happening nearby like harvesting, include these in the walk! Or perhaps it is the time between February and March when women gather to fish in the swamps and small rivers. It is sometimes easier to define seasons according to the farming activities that take place in the area during each season.



3. Talk to the children about where you are going on the walk and what they should look out for. Perhaps each child can collect one or two things, such as a flower, leaf or cone to take back to the playroom or they can collect things, like fallen leaves or seeds, for activities. Take bags to put things in if necessary.

Instructions:

All Ages:

1. Encourage children to look and feel different plants. Discuss how they look - are the leaves green and fleshy or brown, dry and crunchy? Feel the trunks of trees and talk about how they are rough or smooth, hard or soft. If possible, look at the bark through a magnifying glass. You may see small holes through which the tree breathes.
2. Look at the fields and talk about what the children can see and hear. What is growing in the fields? What animals and tiny creatures do they see? Is there water in the dams or streams?
3. Follow up after the walk.
 - Talk about where you went and what you saw during the walk. Draw attention to farming activities and changes related to the weather and the season, e.g. Why is it so dry? Because it does not rain in the dry season.
 - Children can show the group what they collected and talk about it. Things collected can be compared and sorted. Special items can be put in the collection display case.
 - Wild flowers and leaves can be pressed by placing them between the pages of a heavy book (e.g. telephone book). Pile some more weight on the top and leave for a few weeks to dry. Children can then glue them onto cards as a special art activity.
4. Ask the following Questions about what was collected:
 - *What foods will grow from the seeds?*
 - *How does it look / feel / smell?*
 - *What kinds of fruit / vegetables did you see growing?*
 - *What do you think it looks like inside?*
 - *What color is it? Which flowers / leaves / seeds have the same shape /size?*



Clouds

Objectives:

- To observe cloud formations
- To understand that clouds move, change, grow bigger, and produce rain
- To develop imagination

Materials needed: None

Instructions:

All Ages:

1. Have children observe clouds on several days.
2. Ask the children to pretend they are clouds gently moving through the sky, blown by the breeze. Encourage them to explore all the space outdoors.
3. Then they can stand still and take deep breaths, pretending to draw more and more water up from the earth into their bodies. Children draw up their arms as they feel themselves blowing up bigger and bigger.
4. Now they can all move in together to form one big, solid rain cloud.
5. Use two metal objects (drums or pot tops) to make a crashing sound and the rain comes pouring down. The children sink lower and lower as they lose all their rain, and end up relaxed on the ground.



II: PLANTS

BACKGROUND INFORMATION

Plants are different from other living things in that they cannot move and each plant makes its own food, but it does need water, minerals from the soil and sunlight. Children might know already about plants from the theme “Food We Eat”. It provides many opportunities for children to learn more about how plants grow and to gather and analyze data on the many different kinds of plants, especially trees and flowers. It is most important, however, that they learn through observation and discovery by growing plants (gardening) and examining real trees and flowers in their surroundings. Children can also observe local farming activities that happen in each season, and talk about why we need to care for and respect the natural environment.

Following information might be useful for teachers so they can answer children’s questions:

Trees:

- Evergreen forests cover the major part of Liberia – about 235 varieties of trees exist in the country. Some examples are cotton tree, the kola tree and the oil palm. Liberia is known for its rubber trees.
- Among the fruit trees there are Pineapples, citrus fruits, papaya, mango, alligator apple, and avocado.
- Trees are very important. They provide shade and shelter for people, animals and crops. Large forests of trees are important for the weather. They provide food for people (fruits and nuts) and some animals, rubber, cork, timber (wood) for making houses, furniture and many other things, and for fuel (making fires for cooking) and paper which is made from wood pulp.

Grasses belong to the most important plant family economically because they provide the main foods for people and most animals. All grains and cereals such as wheat, maize, rice. Sugar cane and bamboo (reeds) are grasses.

Cassava, cacao, indigo, cotton, upland rice etc. also cover a considerable part of Liberia.

Leaves of plants come in all shapes and sizes:

- Leaves help plants to breathe and live. Each leaf is a tiny factory which uses sunlight, carbon dioxide and water to produce some of the food the plant needs to live. The rest comes through the roots.
- Leaves provide food for animals and people (e.g. leaf vegetables such as cabbage, spinach, lettuce); medicine from herbs; drugs such as tobacco.

Flowers are necessary for plants to reproduce themselves:

- Flower buds are covered in sepals which are usually green but can be bright colors.
- The corolla consists of petals, which have nectar (a sweet liquid) in some plants to attract insects, especially bees. Grass flowers do not have sepals or petals.
- In the middle are the stamens with pollen (male) and the pistil leading to the ovary where the seeds are (female).
- Pollination: insects, bats and the wind help to move the pollen from the stamens of one plant to the pistil of another plant to fertilize the seeds.
- Seed dispersal: the seeds are scattered by the wind (e.g. dandelions, grasses) and by animals (seeds either stick on their coats or in their droppings after being eaten).
- Flowers are used by bees to make honey (from the nectar), as vegetables (e.g. cauliflower, broccoli), for making perfume, and for decoration.

Fruits and **nuts** grow around the seeds and provide food, including vegetables such as pumpkins, tomatoes, cucumber. Cotton and flax (used for making linen) come from seed cases called *bolls*.

Seedless plants have no flowers and consist of a large variety of ferns, which have feathery leaves under which grow *spores* for reproduction (like primitive seeds), and *mosses*.

Other plant-like growths are not true plants and are not green:

- *Fungus* such as mushrooms and toadstools (poisonous) and molds live on other plants (parasites).
- *Lichens* are crusty patches or bushy growths on tree trunks or on the ground.
- *Algae* grow in water (like seaweed) or very damp places



WEEKLY PLANNING OVERVIEW – May, Week 2 and 3

II: Plants

Time	Type of Activity	Monday	Tuesday	Wednesday	Thursday	Friday
9.00 – 9.15	Morning Meeting	• Welcome children • Greetings • Classroom Rules • Trees and Leaves p. 19	• Welcome children • Greetings • Classroom Rules • Leaf Guessing Game p. 18	• Welcome children • Greetings • Classroom Rules • Classification Game p. 21	• Welcome children • Greetings • Classroom Rules • Trees and Leaves p. 19	• Welcome children • Greetings • Classroom Rules • Flowers p. 23
9.15 – 11.00	Work Time – Learning Centers: Teacher organizes materials for children to work in centers of the child's choice including: Art Center, Block Center, Drama Center, Literacy Center, Manipulative Center while one or two Small Groups are working on teacher-initiated activities.					
	Small Groups Week 2	Seed Creatures p. 25	Growing Plants p. 26	Seed Creatures p. 25	Growing Plants p. 26	Bottle Greenhouse p. 27
	Small Groups Week 3	Nature Collage p. 24	Vegetables p. 28	Leaf Patterns p. 30	Vegetables p. 28	Free Play
11.00 – 11.15	Clean-Up					
11.15 – 12.00	Outdoor Time	Flowers p. 23	Young Flowers p. 33	Tree Pictures p. 34	Making Propellers	Free Play
12.00 – 12.45	Wash-Up, Lunch					
12.00 – 12.45	Rest & Quiet Time	Children who can't sleep may "read" books.				
14.15 – 14.30	Group Story Time	Social games, story reading, finger plays, singing, dramatization of the books, reading the children's books related to the topic.				
14.30 – 15.00	Closing Meeting	Review of the day, sharing impressions of the day, discussion of the activities children have been involved in – which new things they learned and planning the next day.				
Activities with Parents throughout the day		• Parents can volunteer in Small Group activities • Parents can join in songs, games, books, storytelling • Parents can accompany class on trash walk				



A. IDEAS FOR THE MORNING MEETING



Leaf Guessing Game

Objectives:

- To develop imagination and creativity.
- To encourage children to ask questions.

Materials needed:

- Cardboard
- Drawings of leaf and caterpillar
- Variety of leaves
- Crayons
- Sand
- Scissors
- Paste/glue

Preparation:

1. Cut out a large leaf shape from the cardboard and color it green.
2. Cut out and color the caterpillar.
3. Paste sand onto the back of the leaf and the caterpillar.
4. Stick the caterpillar onto the flannel board and place the leaf over it so that caterpillar is hidden.

Instructions:

- Beginner:**
- Note: Not for younger children.

**Intermediate/Advanced:**

1. Show the children the leaf and ask them to guess what might be hiding underneath the leaf. Accept all their suggestions and encourage them to think of many different things.
2. Ask some child to come up and lift off the leaf. Talk about why the caterpillar is there and what will happen to the leaf.
3. Give each child a leaf. Tell them to pretend that they have something hidden underneath their leaves. Children take turns to ask questions about the hidden objects: Is it big or small? How many legs does it have? What does it look like? Each child describes what is hidden under his or her leaf in response to the questions and then talks about the reasons for it being there.



Trees and Leaves

Objective:

- To increase awareness of their natural environment.

Materials needed:

- A variety of leaves and other things that grow on trees (fruits, nuts, blossom)

Instructions:**Beginner:**

1. Talk about the uses of trees and leaves (e.g. shelter, shade, wood, paper, rubber as well as medicine and food)
2. Encourage children to look carefully at leaves and talk about their color, texture, shape and smell. Discuss the veins which carry food to the leaves from the roots, and from the leaves (make food) to other parts of the plant. Compare veins in ordinary leave (like a net) and in blades of grass (parallel).

Intermediate/Advanced:

1. Talk about the uses of trees (e.g. shelter, shade, wood, paper, rubber).
2. Encourage children to look carefully at leaves and talk about their color, texture, shape and smell. Discuss the veins which carry food to the leaves from the roots, and from the leaves (make food) to other parts of the plant. Compare veins in ordinary leaf (like a net) and in blades of grass (parallel). Lilies also have long, thin leaves with parallel veins like grass. Ask children if they can find veins on themselves.
3. Children can compare the leaves of different trees. Leaves are also useful for talking about the concept of width (wide and narrow), and children can arrange leaves from widest to narrowest, longest to shortest.
4. Following questions will help you to conduct further discussion:
 - *How does it look/smell/feel/sound?*
 - *What kind of edge does the leaf have?*
 - *Can the leaf be divided evenly in two halves?*
 - *Which leaf is wider/widest; narrower/narrowest?*
 - *Where have you seen one (tree or leaf) like this before?*
 - *What do you think will happen to the bud/flower?*
 - *Why do we need to have trees? What happens if there are no leaves?*
 - *What can we use dry pieces of wood for?*
 - *How do trees help animals/people/other plants?*

Important! If there are poisonous plants in the environment, make sure that children know which ones they are.



Classification Game

Objective:

- To extend children`s understanding of classification and number concepts within the “Plants” theme.

Materials needed:

- Many different leaves or flowers (at least one for each child) for classification activities
- Drawings of leaf shapes and flower types
- Cardboard or paper, scissors, glue

Preparation:

1. The children and their parents can help to collect the leaves and flowers.
2. Make the category cards for the week by cutting out the drawings or pictures needed. Paste these onto separate pieces of cardboard or paper.
3. Set out the category card on a large piece of paper and the items to be classified (e.g. leaves or flowers) flat on the floor or large table. Each child chooses a leaf (or flower) to classify.

Categories: the following ideas for categories are suggested:

1. **Shape of leaves:** oval, round (circle), triangular, and rectangular (long and narrow). Which shape does your leaf fit best? Which shape is your leaf most like? Leaves can also be classified according to size, color, or texture.
2. **Type of flower:** single flower at the end of stem (e.g. rose, arum lily, pansy); composite flower (e.g. daisy, sun-flower); cluster of flowers at top of stem (e.g. geranium); spike of flowers along stem (e.g. grass flowers). Flowers can also be classified according to color, number of petals, shape.

Instructions:

Beginner:

1. Each week set out the selected category cards/activity. Make sure that the children understand what the picture cards show (represent) and know what they have to do (classification activity). Link the task with what they have learned previously.
2. Help children to decide which category to use when necessary.

Intermediate/Advanced:

1. Each week set out the selected category cards/activity. Make sure that the children understand what the picture cards show (represent) and know what they have to do (classification activity). Link the task with what they have learned previously.
2. Help children to decide which category to use when necessary.
3. Talk about how many fall into each category. Ask questions and talk about (compare) the different kinds of leaves and flowers.
 - *Which shape are the most of leaves like? The least?*
 - *Which of the leaves is most like a triangle? Circle?*
 - *Are the leaves/flowers the same color? How many different colors are there? Are all blue flowers exactly the same color?*
 - *How do the leaves/flowers smell/ feel?*



Flowers

Objective:

- To talk about and compare how different flowers look and smell.

Materials needed:

- Flowers of different kinds and colors
- Posters and pictures for different kinds of flowers
- Vases or jars

Instructions:

All Ages:

1. Ask children to bring flowers or to go for a walk to a garden to collect a variety of different flowers. If collected a day before, keep the flowers fresh in a jar of water. In each season there will be different flowers, especially in spring and summer.
2. Ask children to talk about their flowers, how they look (color, shape) and smell. Talk about the different parts of the flower and compare the buds with open flower petals (which contain the nectar that insects like) and the seeds with a weak smell and those with a strong smell) and shape or type of a flower:
 - *How does it look? What color is it?*
 - *How does it smell?*
 - *Why do you think it has long stem/petals?*
 - *Why do you think it is such a bright color?*
 - *What would happen if all flowers were the same color?*
 - *Where have you seen other flowers that are the same?*

B. IDEAS FOR THE SMALL GROUP ACTIVITIES



Nature Collage

Objectives:

- To encourage children to use their imagination and create pictures using leaves/flowers of different colors, shapes and sizes.
- To observe the differences and similarities in the colors, shapes and sizes of leaves and flowers during the dry season.

Materials needed:

- Paper
- Leaves or flowers of different colors, sizes, shapes
- Scissors
- Glue/paste and brushes
- Crayons

Instructions:

All Ages:

1. On a walk, have Children collect leaves from trees or flowers.
2. When they are back in the classroom, have the children create their own pictures by pasting the leaves provided onto paper.
3. The teacher can also provide crayons for children to draw pictures and then fill in these drawings by pasting the leaves on the drawing. Children may also cut up the leaves.



Seed Creatures

Objective:

- To practice small muscle coordination.
- To use one's imagination and develop creativity.

Materials needed:

- Large seeds or pits such as avocado pear or mango (these will need to be cleaned beforehand)
- Variety of material such as twigs, feathers, small seeds,
- Grass, wool and buttons
- Pieces of soft wire or paper clips
- Paste and scissors

Instructions:

All Ages:

1. Children can help to collect the seeds and other materials. They can also be shown how to clean the seeds.
2. Children make seed animals using the above materials.
3. Encourage children to use their own ideas.



Growing Plants

Objective:

- To observe different ways that plants grow.

Materials needed:

- Large glass jar
- Elastic band
- Piece of soft cloth or panty hose,
- Seeds for sprouting

Instructions:

All Ages:

1. Place the seeds in the jar with water. Cover the jar with the cloth and elastic band.
2. Soak the seeds overnight and drain the water the next day through the cloth.
3. Rinse the seeds daily and drain.
4. After 4-6 days, the jar will be full of sprouts. Let the children eat them raw or cook them slightly (they are full of vitamins).



Bottle Greenhouse

Objective:

- To make a greenhouse.
- To watch plants grow from seeds.

Materials needed:

- 2 liters plastic Coca-Cola bottle
- Scissors or knife, hammer and nail
- Soil, seeds (flower or vegetable seed)

Instructions:

All Ages:

1. Twist off the base of the bottle.
2. Make drainage holes in the centre of the base with the hammer and nail.
3. Fill the base with soil. Place the seeds in the soil. Water well.
4. Cut the top part of the bottle just at the point where the angle of the bottle becomes straight.
5. Use the neck part of the bottle for a water or sand play funnel.
6. Place the long part of the bottle over the base bottom up. It will fit snugly and provide warmth like a greenhouse. After the dome is on, it will not need watering again.



Vegetables

Objective:

- To observe how different vegetables and fruits grow from different beginnings.

Materials needed:

- Ginger root
- Butter pear
- Avocado pits
- Onions/potatoes that have sprouted

Instructions:

1. Plant a piece of ginger root by placing it on the surface of the soil in a shallow container. Roots will grow on the underside. A wonderful ginger-scented plant will grow.
2. Suspend an avocado pit in a jar of water using match sticks.
3. Do not throw away sprouting onions and potatoes. Plant them out in the garden or in a container.



Flower Pot Game

Objective:

- To encourage concentration and cooperation skills.

Materials needed:

- 4 small cups and one larger container
- Thin cardboard (e.g. from cereal boxes)
- 24 straws or sticks
- Scissors, paint, crayons, paste or tape

Preparation:

1. Paint each cup a different color (use colors that would be found on flower petals, e.g. red yellow purple, pink)
2. Cut out the flower shape. Stick it onto cardboard and then use it as a stencil to make 24 petal shapes on thin cardboard.
3. Paint or color in the flowers so that there are 6 of each color to match the yoghurt containers.
4. Make a dice and write the numbers 0, 1, 2, 0, 1, 2 on each side.

Instructions:

All Ages:

1. Four children can play the game. Put all the towels in a container in the center of the playing area. Explain the rules of the game and how to play.
2. Children take turns to throw the dice. They collect that number of flowers to match the color of their yoghurt cups. If a 0 is thrown, the child misses a turn. The game continues until all the flowers are sorted.



Leaf Patterns

Objective:

- To learn about numbers and patterns.

Materials needed:

- Variety of leaves that differ in shape and size
- Cardboard, paste and brush
- Plastic or clear contact
- Box for storage

Preparation:

1. Press the leaves in a heavy book for a few weeks.
2. Cut the cardboard into 10 cards of the same size (about 10 cm x 30 cm).
3. On each card paste leaves in different patterns according to the shape and size.
4. Cover the cards with plastic or clear contact.
5. Put leaves that are the same as those on the cards in a storage box.

Instructions:

All Ages:

1. The child selects a card and chooses leaves according to the pattern and places them underneath the card.



Matching Seeds

Objective:

- To develop perceptual discrimination in matching the seeds.

Materials needed:

- 1 cardboard egg box (12)
- Scissors, crayons
- 12 pairs of seeds or beans (e.g. pumpkin, squash, watermelon, orange, apple)

Preparation:

1. Cut off the lid of the egg box, then decorate the bottom half with the crayons. This is the matching tray.
2. Put all 24 seeds in the lid of the egg box

Instructions:

BEGINNER:

1. Children try to find a pair of seeds that are the same for each cup.
2. Extend the experience by matching the seeds or beans to 12 corresponding drawings.

INTERMEDIATE/ADVANCED:

1. Children try to find a pair of seeds that are the same for each cup.
2. Extend the experience by matching the seeds or beans to 12 corresponding drawings.
3. Have the children sort leaves according to their shapes and colors (autumn).

C. IDEAS FOR OUTDOOR ACTIVITIES



Flowers

Objective:

- To learn how plants grow from seeds.
- To learn what plants need to stay alive and healthy.

Materials needed:

- Flower seeds
- Good soil
- Planter (if necessary) – tractor tire, old tin bath, or any other large container

Instructions:

1. Make a flower garden with the children outside or in small sections of the playground where the children do not need to play. If there is a flower garden already, encourage the children to help preparing the soil, planting and looking after the plants (watering, weeding). Plant a variety of flowering plants at different times of the year so that there will be flowers in every season. Remember that some plants like sun and others like some shade or protection from wind.
2. If space is limited or the soil is unsuitable, flowering plants can be grown in tubs made by filling a large tire (preferably a tractor tire), an old tin bath or any large container with good soil. If the container has a solid bottom makes holes and put a layer of stones at the bottom to provide drainage. These containers can also be placed on a verandah if there is enough light and sun.
3. Flowering plants can also be grown in pots or tins inside. Choose plants suitable for indoor conditions make sure there are holes for drainage and use good soil. Explain to the children how much water and how often to water - pot plants should not be over-watered.
4. Make a duty roster for the older children to take turns in looking after the plants.



Young Flowers

Objective:

- To nurture imagination and creativity.

Materials needed:

- None

Instructions:

1. Let the children pretend to be new young flowers, pushing up through the earth. They begin sitting cross-legged on the floor, heads down.
2. Place palms of hands together and push them above their heads. Gradually stand and unfold their arms like flower petals.
3. Now pretend a wind has come up, making the flowers sway and swing. Here comes a stronger wind with a whoosh, whirling the flowers right round and blowing them all over the room.
4. Away goes the wind and the children gently settle down on the floor for a rest..



Tree Pictures

Objective:

- To nurture imagination and creativity.

Materials needed:

- None

Instructions:

1. Divide the children into groups of 3 or 4. Together they show what different trees look like, e.g. they show what a tall thin tree looks like by standing close together with their arms stretched up above their heads and hands together.
2. Ask the children to show what a round bush looks like; then trees that are flat at the bottom and come to a point at the top like a triangle; then trees that are flat on top with big branches going outwards; and so on.
3. Choose trees that the children are familiar with. They can also suggest the kind of trees or other plants that they can make.



III: ANIMALS: FARM ANIMALS AND PETS

BACKGROUND INFORMATION

In the animal world there is a large variety also – in the forests you can find Bosman's potto that is in fact a lemur. Monkeys of numerous species are also found in Liberia. Earlier buffalo and elephant were in great numbers in Liberia, but these have disappeared nowadays. Wild pigs and porcupines along with some categories of leopard are also there in some parts of Liberia. Some species of antelope are still found in the forests of Liberia.

However, this topic is about domestic animals, which means those animals that are kept by people as pets and for the economic activity of farming. These are the animals like dogs and cats which are usually kept by families as pets but are often useful on farms as well. It can also include horses, donkeys, mules, cattle, sheep, goats and pigs; poultry such as chicken, turkeys, ducks and geese; ostriches, and even fish such as trout.

Younger children (2-4 years) need to see and talk about real animals. They should also learn about one kind of animal at a time. Older children (4-6 years) can learn more by comparing the different animals and talking about the ways they are the same and different.

Children can also observe how different animals move, and hear the sounds they make. They can learn about the care of the animals, what they eat, and the names of the males, females and young ones.

It is important for children to know that farm animals are **living things**, like people: they need food and water, they grow and reproduce, they feel pain when badly treated, they can get sick and need medicine, and they need shelter from bad weather and protection from wild animals.



WEEKLY PLANNING OVERVIEW – May, Week 4; June, Week 1

III: Animals: Farm Animals and Pets

Time	Type of Activity	Monday	Tuesday	Wednesday	Thursday	Friday
9.00 – 9.15	Morning Meeting	- Welcome children - Greetings - Classroom Rules - Learning About Domestic Animals p. 37	- Welcome children - Greetings - Classroom Rules - Different Kinds of Animals p. 39	- Welcome children - Greetings - Classroom Rules - Learning About Domestic Animals p. 37	- Welcome children - Greetings - Classroom Rules - Different Kinds of Animals p. 39	- Welcome children - Greetings - Classroom Rules - Learn About Pets: Dogs p. 41
9.15 – 11.00	Work Time – Learning Centers: Teacher organizes materials for children to work in centers of the child's choice including: Art Center, Block Center, Drama Center, Literacy Center, Manipulative Center while one or two Small Groups are working on teacher-initiated activities.					
	Small Groups Week 4	Making Farm Animals & Pets p. 47	Animal Masks p. 49	Making Farm Animals & Pets p. 47	Animal Masks p. 49	Making Butter p. 52
	Small Groups Week 1	Egg Shell Art p. 42	Collage Ideas p. 46	Egg Shell Art p. 42	Collage Ideas p. 46	Heads & Tails p. 55
11.00 – 11.15	Clean-Up					
11.15 – 12.00	Outdoor Time	Making Butter p. 52	Heads & Tails p. 55	Animal Movements & Sounds p. 57		Free Play
12.00 – 12.45	Wash-Up, Lunch					
12.00 – 12.45	Rest & Quiet Time	Children who can't sleep may "read" books.				
14.15 – 14.30	Group Story Time	- Social games, story reading, finger plays, singing, dramatization of the books, reading the children's books related to the topic. - Discovery Box: exploring sounds, tastes, smells, things found in nature and manufactured - Movement game "Touch blue..." - Asking questions about what did children find during Sense Walk				
14.30 – 15.00	Closing Meeting	Review of the day, sharing impressions of the day, discussion of the activities children have been involved in – which new things they learned and planning the next day.				
Activities with Parents throughout the day		- Parents can volunteer in Small Group activities - Parents can join in songs, games, books, storytelling - Parents can go with class on a Community Walk				



A. IDEAS FOR THE MORNING MEETING



Learning About Domestic Animals

Objectives:

- To familiarize children with the variety of domestic animals.
- To begin categorization: farm animals and pets.

Materials needed:

- Pictures of typical farm animals and pets (if available)

Preparation:

1. Plan each talk-about time to focus on a particular topic. The age of the children is so important in planning around these topics that two sets of talk-about ideas are suggested.
2. With young children (2-4 years) talk about only one or two kinds of farm animals or pets at a time. Begin with the most common animals, such as dogs and cats, which children will see around their homes. Only talk about animals that children can actually see on their farms.
3. **Older** children (4-6 year-olds) can compare different farm animals and talk about different types of stock farming. They can also learn about animals and farming activities that may not be familiar to them. Some of the talk-about ideas suggested for older children are sequenced to help children learn more about animals and develop their thinking and language skills.
4. Try to use a talk-about topic in group time about twice a week. Do not try to discuss too many different ideas in one Group Time.
5. Where possible link the talk-about topics with related special activities, stories, music and movement. Try to build on what the children have learned or experienced before. This helps children learn.

6. Begin each talk-about time with a suitable game, movement activity, rhyme or song to involve all the children actively in the group time.

Visual aids:

- Suggested posters and drawings are illustrated in the text. Also use suitable magazine pictures and other posters where possible.

Instructions:

All Ages:

1. Have the children see and talk about real animals. Either bring animals - a dog, cat or hen, to the school or take the children for a walk to look at animals if they are nearby. The children will also enjoy seeing young animals, such as a cat with kittens, chickens with chicks, sheep with lambs.
2. If the children go out to see animals, use pictures (drawings or magazine pictures) when you talk about the animals during the Group Time.
3. If it is not possible for children to see real animals, then use pictures and models in clay or plastic. Also have real things associated with the animals, e.g. feathers and eggs when talking about fowls; milk from the cow; sheep's wool. Do NOT talk about animals that most of the children do not see.
4. In the last review session, use pictures of the all the animals the children have talked about.
5. Things to talk about each animal:
 - *What it is called (name).*
 - *What it looks like (appearance) and the different parts of its body; what the young animals look like,*
 - *How it moves and the sounds it makes.*
 - *What it eats and what kind of shelter it needs.*
 - *The uses or products of the animals, e.g. cats catch mice; leather for shoes, belts, bags, bridles; meat, milk, eggs for food.*



Different Kinds of Animals

Objective:

- To find out what children know about different kinds of domestic animals, including the names of male, female and young animals, and the sounds they make.
- To compare the appearance of different animals and the way they move.

Materials needed:

- Make posters of different farm animals and their babies using available drawings and magazine pictures.

Instructions:

All Ages:

1. Talk about each kind of animal using the posters. Ask them what the mother, father and babies of each of the animals is called. Children can learn the names of the sounds they make, e.g. *bark, low, bleat, cluck*.
2. Talk about what the animals look like and the ways in which they are the same and different. Ask questions about their body parts, skin covering, color, size, and so on.
3. The children can imitate the way different animals move. Use the words that describe the movements, e.g. *gallop, trot, climb, waddle, and paddle*.
4. Ask the following questions
 - *What color are cows/horses/goats/pigs/hens/ baby chickens? Which animals are the same colors?*
 - *Which animals have long legs? And which have short legs? Why do you think they have long or short legs?*
 - *Which animals have 4 legs, and which have 2 legs?*
 - *Which animals have horns? Do the horns look the same? What do horns look like? What do they feel like?*
 - *Which animals have a comb, udder, webbed feet, curly hair, wool, claws; chew the cud, long ears?*
 - *Are all the animals' ears, mouths and eyes the same size and shape? Which animal is the biggest?*
 - *How do the horses and cows move? Is this different from the way the ducks and chickens move? Why?*



Where Does It Belong?

Objective:

- To match different animal part to the animals they belong to.

Materials needed:

- One drawing of each kind of farm animal
- Drawings of noses, tails, feet
- Cardboard, scissors, crayons/paint, paste

Preparation:

1. Color in (or paint) and cut out each animal separately. Stick it onto a piece of cardboard.
2. Color in and cut out the drawings of different animal parts. Stick each page onto pieces of cardboard. When the glue is dry, cut the card with animal parts into the smaller cards.

Instructions:

All Ages:

1. This talk-about time could be linked with a walk to look for different animal footprints in the ground.
2. Begin with a guessing game, e.g. I am big and strong; I have four legs, a mane and a long tail; people ride on my back. What am I? I have 4 legs, big eyes, a long tail to whisk away flies, and an udder full of milk?
3. The children sit in a circle. Arrange the larger cards with the pictures of different animals in a circle in front of the children on the floor. Place the smaller cards with the animal parts face down in the middle of the circle.
4. Each child has a turn to select a card and place it next to the picture of the animal where it belongs, e.g. curly tail on pig, webbed feet on duck.
5. Talk to children about the differences between these body parts of animals. Ask the children about which animals have the same feet, tails or noses.



- *Why does a duck have webbed feet but not a chicken?*
- *What is the duck able to do with its webbed feet?*
- *What is the difference between the hooves of horses and those of cows? What other animals have cloven hooves like cows?*
- *Why do cats and dogs have claws on their paws?*
- *What if none of the animals had tails? Why do animals need to have tails?*
- *What is the difference between the horns of the cow and those of the goat? What other animals have horns?*
- *Why do poultry have beaks? Are the beaks all the same? What is the beak of a duck called? Do chickens have teeth?*



Learning about Pets: Dogs

Objectives:

- To practice pre-math skills.
- To learn names of body parts.
- To compare and contrast.

Materials:

- A calm and friendly dog

Instructions:

All Ages:

1. Ask the children whether they have dogs at home. They probably talked about dogs in ***My Family***, so encourage them to do most of the talking by asking suitable questions.
 - *How many ears/eyes/nose does the dog have?*
 - *How many ears do you have? Touch them.*
 - *Do your ears look the same as the dog's ears? Why not?*
 - *What do dogs do with their paws? (scratch, dig)*
 - *Pele, is your dog at home as big as this dog (name)?*
 - *Why do people keep dogs? Do dogs help on the farm?*
2. Tell them the name of the visiting dog. Let one or two children at a time pat or stroke the dog. While the others watch, talk about the appearance and care of dogs.
3. Children can point out the different parts of the dog's body, e.g. head (talk about the ears, eyes, nose, mouth, tongue, teeth), neck, back, stomach, tail, legs (count), paws and claws, and the hairy coat (what does it feel like). Encourage the children to talk about how the dog's body is the same or different from themselves.
4. Children can also talk about the differences between the visiting dog and their own dogs, e.g. in size, whether the ears and tails are the same, the hair and the color.
5. The children can try to walk and run like dogs on four legs. They can make the sounds that dogs make (barking and whining).
6. Talk about what dogs eat and how they are cared for. They need food, water, shelter and exercise because they are living things, just the same as people. And just the same as children, they do not like to be hurt.



Animal's Food

Objective:

- To group together (classify) food products and things connected with different kinds of animals.

Materials needed:

- Drawings and magazine pictures of food products that we get from farm animals and other things connected with animals
- Large drawings of animals (horse, cow, sheep, pig, chicken and duck)
- Cardboard (from cereal boxes), scissors, paste
- Crayons and paint

Preparation:

1. Color in the drawings. Cut out the pictures and paste each one onto a card about 6 or 8 cm square.
2. Put the smaller picture cards into a box or bag.

Instructions:

BEGINNER:

1. Arrange the bigger animal pictures in a row next to each other either on the floor or on the wall where all the children can see them and reach them.
2. Each child takes a small picture from the bag or box and has a turn to stick it underneath the farm animal that it comes from or is connected with, e.g. milk and leather goods underneath the cow, woolen things under the sheep.
3. First ask the child to say what the picture is of. If needed, help the child decide where to put the card by asking questions. Ask the child to say why he has put the picture with a particular animal, and then ask the group if they agree with where the picture is placed. If not, why they disagree.

INTERMEDIATE/ADVANCED:

1. Arrange the bigger animal pictures in a row next to each other either on the floor or on the wall where all the children can see them and reach them.
2. Each child takes a small picture from the bag or box and has a turn to stick it underneath the farm animal that it comes from or is connected with, e.g. milk and leather goods underneath the cow, woolen things under the sheep.
3. First ask the child to say what the picture is of. If needed, help the child decide where to put the card by asking questions. Ask the child to say why he has put the picture with a particular animal, and then ask the group if they agree with where the picture is placed. If not, why they disagree.
4. Talk about the things, especially leather goods and meat products, which can come from different kinds of animals.
5. When all the pictures have been sorted, count how many products we get from each animal. Which is the least? The most?
6. Ask questions to help the children think of other ways of classifying the products we get from animals, e.g. food, leather goods, clothing, bedding; or things from live animals and things from dead animals. Decide on the categories, and let the children sort the pictures again.
 - *Do we eat duck's eggs, and duck meat? What are the duck's eggs used for? (breeding more ducks)*
 - *What other animals' skins are used to make leather goods? (pigs, sheep)*
 - *Which animals give us meat to eat? Does the meat from a chicken and a pig look the same?*
 - *Are cows the only animals that give us milk to drink?*
 - *What about goats?*

Extensions:

1. If you can find enough pictures of animals, children can sort these in different ways depending on the selection of pictures, e.g.
 - Different kinds of farm animals
 - Young animals and adult animals
 - Different color animals
 - Large and small animals
 - Animals with 4 legs and 2 legs
2. Children could also sort pictures of different breeds of only one kind of animal, e.g. cattle or dogs or cats, if you happen to find many pictures of these.
3. Set out the different sets of pictures in suitable containers with sorting trays during free choice activity time for children to sort individually or in pairs.



B. IDEAS FOR THE SMALL GROUP ACTIVITIES



Eggshell Art

Objectives:

- To encourage creativity and imagination.
- To practice small muscle skills and eye-hand coordination.

Materials needed:

- Empty eggshells (washed)
- Paints made with mud or dyes or poster paints and
- Brushes
- Sharp object (pin, knife, scissors) and thread (e.g. wool, cotton, string), sticks or wire coat hangers
- Rollers (bottles, tins) and paste, paper

Preparation:

1. Plan this activity well in advance to be able to collect enough eggshells. Perhaps ask children to bring eggshells from home. Make sure that the eggshells are washed out to avoid unpleasant smells.
2. There are several different activities that can be done on different days.

Instructions:

All Ages:

1. Painting:

- Put the eggshells out with the paints and brushes.
- Encourage the children to paint patterns on the eggshells using different colors (if available).
- Leave the eggshells to dry in a safe place.

2. Threading:

- When the eggshells are dry, very carefully make a small hole in each shell with a sharp object.
- Show the children how to thread the shells onto a piece of string or wool to make mobiles. Knot the string or wool after threading about 4 or 5 shells.
- Tie two sticks across each other. Then help the children to tie the threaded eggshells to the sticks (or wire coat hangers), and hang the mobiles up in the playroom.

3. Alternative:

- Break the eggshells into smaller pieces and make a hole in each piece.
- The children thread the pieces onto a piece of thread to make bangles or necklaces.

4. Collage:

- Show the children how to crush the painted eggshells into very small bits with rollers and put them in a container.
- Encourage the children to make pictures by pasting the crushed eggshells onto paper.



Making Farm Animals and Pets

Objectives:

- To encourage children to use their small muscles to mold, pinch and create farm animals.
- To encourage creativity and problem-solving.

Materials needed:

- Clay or play dough or papier-mâché or thick soft roots.
- Collections of seeds, leaves, sticks and small stones,
- Small feathers

Instructions:

All Ages:

1. Arrange the collections of seeds, leaves, sticks, small stones, and feathers attractively in separate containers on the table or floor where children will work.
2. Set out the clay, play dough or papier-mâché in the usual way at each place where children will sit. If not available, put the thick roots or whatever you can find that children could use to make animals with.
3. Encourage the children to mold the clay or play-dough or papier-mâché. If necessary, show children how clay can be pinched, smoothed, rolled or how to work with the papier-mâché.
4. Ask the children questions that will encourage them to use the seeds, sticks, feathers and other natural materials for legs, ears, wings, and so on, e.g. *What could you use to make the cow's horns?*
5. If modeling material (like clay) is not available, talk to the children about how they could use the natural materials to make animals by using the thick roots for the body, sticks for legs, and so on.

NOTE: Do not tell children to make specific animals or give them models to copy. Encourage them to be creative.



Animal Masks

Objectives:

- To make animal faces which can be used in movement and story time.
- To encourage observation of details and develop memory skills.

Materials needed:

- Pieces of stiff paper or thin cardboard
- Crayons, scissors, glue, brushes
- Pieces of cardboard in different shapes for ears, horns
- String or thick wool

Instructions:

All Ages:

1. Encourage children to make animal faces or masks that they can wear. Ask them to think about the kind of animal they would like to be, and then draw the face onto a piece of stiff paper and cut it out. Help them to cut holes for their eyes.
2. Ask them to use the cardboard shapes to stick on ears and horns or make a mane with bits of wad, depending on the kind of animal.
3. Help them to tie pieces of string or wool onto their masks so that the masks can be tied around their heads.

Alternative:

1. Glue the masks onto headbands made out of strips of thin cardboard or folded newspaper about 50 cm long. Help them to measure how long a strip they need to go around their heads and stick it to make a band. If necessary, help them to stick the faces with all the different pieces onto the headbands. Allow the glue to dry before they wear the masks.
2. Clearly mark each child's mask with his\her name or symbol.



Collage Ideas

Objectives:

- To encourage creative sticking activities using animal products.

Materials Needed:

- Scraps of knitting wool in different colors
- Feathers
- Sorting trays, boxes
- Paper and paste

Instructions:

All Ages:

1. Set out special collage activities which encourage children to make pictures and patterns with one type of material on different days.
 - **Wool:**
 1. Children can sort the scraps of wool (cut up long pieces) according to the different colors in a sorting tray. Place the sorting tray, paste containers and paper on the floor or table where the children will work.
 2. Let the children make whatever pictures or patterns they like by pasting the bits of wool onto a piece of paper.
 - **Feathers:**
 1. Children could also sort the feathers first into different sizes, shapes or colors.
 2. They stick feathers onto the paper in whatever patterns they like.



Cooking Eggs

Objectives:

- To discover what happens to eggs when they are cooked in different ways.
- Cooking activities should be supervised at all times. Make sure that the children clearly understand the rules, e.g. how far away they should be from the stove.

Materials Needed:

- One egg per child
- One fork or spoon per child (for mixing)
- Large spoon, knife, egg lifter
- A form of heating (stove, gas burner, primus, fire)
- One empty container per child (e.g. small margarine containers)
- Saucepan or pot, frying pan

Instructions:

1. Work with a small group of not more than six older children at one time so that they can be carefully supervised and helped where necessary. Other children can watch if they wish, and everyone should have some to eat afterwards. A different group can take part in each of the suggested discovery activities.
1. Set up the activity on a low table near the stove. There should be enough space for the children to sit or stand around the table comfortably.
 - **Scrambling eggs:**
 1. Give each child a container, fork or spoon, and one egg.
 2. Show the children how to gently crack the egg by knocking it against the side of the container, and then empty the contents into the container.
 3. Talk about what a raw egg looks like: the clear color of the white and the yellow of the yolk.
 4. Show the children how to mix or whisk the eggs using their forks. Discuss this as they go along: What does it feel like? What color do you have now?



5. Melt a little butter in the pan. Children then empty their mixed eggs into the pan (add a little water or some milk to make a bigger amount).
 6. While the mixture is cooking the children can take turns to stir it so that they feel and see it getting thicker. Also remove the pan from the heat from time to time for the children to see the changes in the egg.
 7. When the scrambled egg is cooked, divide it up among the children so that they can look at it, feel it and eat it. Talk about the changes: What did the raw egg look like? In what ways is the cooked egg different? What color is it? What does it taste like?
- **Boiling eggs:**
 1. Talk about the fact that the egg is placed into boiling water with the shell still on, the hardness of the shell, color, and shape (oval).
 2. When the eggs have boiled hard, remove them from the boiling water and put them in cold water to cool down. Each child peels the shell off an egg. Talk about how the egg feels, the color and shape.
 3. Encourage children to cut the egg in half and look at the yolk and white. Give each child a piece of hard-boiled egg to eat. Ask them to compare the taste of scrambled and boiled eggs.
 - **Frying eggs:**
 1. A small group of children can observe the changes as an egg is being fried. Ask them to describe what is happening to the egg. They can compare its taste and feel with the scrambled and hard-boiled eggs.

*****Remember to collect the egg shells and wash them for Eggshell Art activities.*****



Making Butter

Objectives:

- To discover how to make butter from milk (cream).

Materials Needed:

- Cream **or** milk with high butter (fat) content
- Small jars or containers with lids
- Salt
- Fine strainer
- Larger bowl
- Wooden spoon

Instructions:

All Ages:

1. If cream is not available, cover a jug or jar of milk and allow it to stand overnight in a cool place. The next morning carefully remove the layer of cream on top of the milk with a spoon and place it in a clean jar. This process can **be** repeated the next night to collect enough cream. Some children may want to observe this stage of separating the cream from the milk.
2. **Step 1:** Half fill the small jars (purity jars are ideal) with cream and screw the lids on tightly. Let children take turns (one jar between two children) shaking until the cream separates.
3. Talk to the children about how it looks and feels: at first it looks like whipped cream, then it becomes thick and lumpy, and finally small lumps of yellow butter separate from a milky liquid (called buttermilk).
4. **Step 2:** Tell the children to pour off the buttermilk through a strainer and taste it.
5. Scrape all the butter bits into the larger bowl and show the children how to work the bits together with the wooden spoon while rinsing the butter in water a few times until the water is clear. The children can take turns working the butter with the wooden spoon. Add a pinch of salt.
6. If bread is available the children can spread the butter on the bread, divide it up and eat it.



Happy Animals Family

Objectives:

- To develop memory skills, concentration and strategic-thinking skills.
- To learn to follow the rules of a game..

Materials Needed:

- Pieces of cardboard (e.g. from cereal boxes)
- Animal family drawings (15 cards)
- Paint or crayons
- Scissors
- Paste/glue
- Storage box

Preparation:

1. Color in the drawings of the animals and stick them onto the pieces of cardboard.
2. Cut out the 15 cards. Make sure that all the edges are stuck down, and if possible cover with plastic contact or clear varnish.

Instructions:

BEGINNER:

1. Give the animal family drawings to children and ask them to sort the cards into the five families. Talk to them about the different animals and what they are called, what they look like, what they eat, and so on.

INTERMEDIATE/ADVANCED:

1. Introduce the game to the children. Show them all the cards and talk about the five animal families. Make sure they know the names of each animal. Explain the rules of the game and see that they are followed.
2. 5 children can play at a time. Mix up (shuffle) the playing cards and give (deal) each child one card at a time until

they each have 3 cards (place the cards face down in front of each child).

3. The children pick up their cards so that the others cannot see what they have got. The aim of the game is for each child to collect one animal family.
4. The first child (sitting next to the dealer) starts the game by asking one of the other children for a card to go with one of the cards he already has, e.g. *Michael can you give me the hen, please*. If Michael has the hen card, she has to hand it over, and the first child gives Michael any one of his other cards (face down so the others cannot see).
5. The child sitting next to the first child then has a turn to ask someone for a card, and so on.
6. When a player has collected an animal family, she puts her 3 cards face up on the table for the others to see that they are correct. The remaining players continue the game until all the families are complete.
7. When the children have learned how to play the game, they can continue playing on their own.



Heads and Tails

Objectives:

- To learn more about the body parts of different animals, and about part-whole relationships.
- To encourage children to pay attention to details and develop their observational and memory skills.

Materials Needed:

- Pieces of cardboard (e.g. from cereal boxes)
- Farm animal drawings (12)
- Paint or crayons
- Scissors
- Paste
- Storage box

Preparation:

1. Color in the drawings of the animals and stick them onto the pieces of cardboard.
2. Cut out the 12 animal cards and then cut each through the middle on the dotted lines. Make sure that all the edges are stuck down, and if possible cover with plastic contact or clear varnish.

Instructions:

BEGINNER:

1. This game can be played individually or by 2 to 4 children as a group.
2. Spread out the 24 cards face up on a table or floor, and explain to them how to match the head to the tail of each animal. For example, point to the horse's head and ask the child(ren) what animal this is. Then ask her to find the horse's tail. Then ask a child to choose a card and tell you what it is. Then he looks for the other half.

INTERMEDIATE/ADVANCED:

1. This game can be played individually or by 2 to 4 children as a group.
2. First look at the cards together and talk about the different animal parts, e.g. what they look like, similarities and differences, why they think they are different, e.g. webbed feet as opposed to chicken feet, paws, hooves, and so on.
3. Mix up the 24 cards and spread them out face down on a table or on the floor. The first child turns face up any two cards. If they match, she takes them, and she turns up another two cards. If the cards do not match, the same child turns them face down again in exactly the same places.
4. The next child then has a chance to turn over any two cards. As the children see the different cards being turned up, they can remember where a particular card is and this helps them to find the other half.
5. They continue until all the pairs have been found.
6. The game can be made simpler at the beginning by using fewer cards, e.g. only 8 pairs (16 cards).



C. IDEAS FOR OUTDOOR ACTIVITIES



Animal Movements and Sounds

Objectives:

- To learn to categorize.
- To stimulate their imagination.
- To increase control of large muscle movement.
- To practice sounds of different animals.

Materials needed:

- One Animal Mask for each child

Instructions:

BEGINNER:

Note: Not for younger children

INTERMEDIATE: ADVANCED:

1. Start off with a finger play or rhyme to settle children and get their attention.
2. Ask children to put on the animal masks that they made, and then divide into animal groups, e.g. sheep, horses, cows, chickens. Do not have more than 5 or 6 children in a group.
3. Each group sits in a different part of the space. The groups take turns to show the rest of the children all the different movements their animals can make, e.g. the horses gallop, trot, walk, jump.

4. Encourage the children to move freely and use their imagination. Only if you find that they are not sure of movements, and then start off by showing one or two yourself. Encourage them to do movements standing up and on all fours.
5. When the group has shown their movements, all the children to copy the movements.

Sounds:

1. As they move around the room making the movements, they also make the animal noises. Start off with one group at a time, and then all the groups move around together making the different sounds.
2. End the group activity with each animal group going to sleep in their different areas of the room, e.g. horses in stables, sheep in the kraal, and cows in the shed.



IV: ANIMALS: SMALL ANIMALS

BACKGROUND INFORMATION

This topic about small animals focuses on small wild animals that live nearby in the back yard or garden, in sheds, in the veld and in or near rivers and dams. The following kinds of animals are included: small mammals such as mice and hares, birds, reptiles, amphibians (frogs), and fish. Separate topics are about smaller creatures such as insects, worms and snails.

The previous topic “Farm Animals” helped children learn more about domestic animals which are mainly mammals. This topic extends children’s knowledge of mammals, birds and fish, and introduces new types of animals, reptiles and amphibians which can live in water and on land.

The purpose is not to make children memorize a lot of facts about specific kinds of small animals, but to arouse children’s curiosity about nature so that they become more observant in their everyday lives. Teachers can do it by creating opportunities for observing and talking about particular animals in small and large group activities.





WEEKLY PLANNING OVERVIEW – June, Week 2 and 3

IV: Animals: Small Animals

Time	Type of Activity	Monday	Tuesday	Wednesday	Thursday	Friday
9.00 – 9.15	Morning Meeting	- Welcome children - Greetings - What is the Topic of the week? - Small Animals and Reptiles p. 61	- Welcome children - Greetings - What is the Topic of the week? - Animal Homes Classification p. 64	- Welcome children - Greetings - Discovering About Birds p. 65	- Welcome children - Greetings - Animal Movements & Sounds p. 66	- Welcome children - Greetings - Animal Homes Classification p. 64
9.15 – 11.00	Work Time – Learning Centers: Teacher organizes materials for children to work in centers of the child's choice including: Art Center, Block Center, Drama Center, Literacy Center, Manipulative Center while one or two Small Groups are working on teacher-initiated activities.					
	Small Groups Week 2	Group Nature Collage p. 68	String Painting p. 69	Group Nature Collage p. 68	String Painting p. 69	Free Play
	Small Groups Week 3	Spot the Small Animal p. 72	Observing Small Animals p. 70	Spot the Small Animal p. 72	Observing Small Animals p. 70	Discovery Table p. 74
11.00 – 11.15	Clean-Up					
11.15 – 12.00	Outdoor Time	Nature Walk p. 75	Catch the Snake's Tail p. 77	Nature Walk p. 75	Catch the Snake's Tail p. 75	Free Play
12.00 – 12.45	Wash-Up, Lunch					
12.00 – 12.45	Rest & Quiet Time	Children who can't sleep may "read" books.				
14.15 – 14.30	Group Story Time	- Social games, story reading, finger plays, singing, dramatization of the books, reading the children's books related to the topic. - Discovery Box: exploring sounds, tastes, smells, things found in nature and manufactured - Movement game "Touch blue..." - Asking questions about what did children find during Sense Walk				
14.30 – 15.00	Closing Meeting	Review of the day, sharing impressions of the day, discussion of the activities children have been involved in – which new things they learned and planning the next day.				
Activities with Parents throughout the day		- Parents can assist in visiting the classroom and telling about their job/role - Parents can assist children in making their postcards - Parents can assist as volunteers in a field trip				



A. IDEAS FOR THE MORNING MEETING



Small Mammals and Reptiles

Objectives:

- To talk about and compare small mammals and reptiles.

Materials needed:

- Drawings and pictures of small mammals and reptiles
- Flannel board, easel, and blackboard or wall space
- Live animals, such as a tortoise or chameleon (if available)

Preparation:

Color in available drawings of small animals and reptiles. Cut out all available pictures and paste them onto cardboard. If using a flannel board, put a strip of paste on the back of the pictures and shake sand onto it so that they will stick to the flannel board.

Instructions:

BEGINNER:

1. Plan several group times on different animals. In each talk about only one or two animals that children have actually seen.
2. If there are any live animals in the playground or garden, take the children to look at them and talk about them. Then return to the playroom and sit down.
3. Tell the children that you are going to talk about small animals that live on or in the ground. Build on the experience of the walk and ask the children to name as many of these kinds of animals as they can. Put up the pictures of one of the animals that they saw or that you have chosen for the day.

4. Ask them questions that are of interest/discuss the animal:

Things to talk about:

- *What it is called (its name).*
- *What it looks like (appearance) and the different parts of its body; what the body is covered with; what the young animals look like.*
- *How it moves and whether it moves fast or slowly.*
- *The sounds it makes; high or low, loud or soft.*
- *How dangerous it is; how it can hurt people.*
- *What it eats and when (some animals only eat at night).*
- *Where it lives or how it is protected; whether it is warm-blooded or cold-blooded.*
- *How babies are produced (live birth or eggs) and how they are cared for.*

5. *Additional Questions*

- *Where have you seen one like this before? How did it look? How does it move?*
- *How do you think its body feels?*
 - *Why do you think it has...(e.g. such a long tongue / no legs / a shell on its back / quills)? What do you think it eats?*
 - *Which of these animals live in holes in the ground?*
 - *Where do the others live?*
 - *In what ways are these animals the same? (e.g. most of them have 4 legs; small mammals have fur and give birth to their babies; reptiles have scales and lay eggs) Which animals move fast and which move slowly?*

INTERMEDIATE/ADVANCED:

1. Begin with a guessing game or movement activity and/or song.
2. If there are any live animals in the playground or garden, take the children to look at them and talk about them. Then return to the playroom and sit down.
3. Tell the children that you are going to talk about small animals that live on or in the ground. Build on the experience of the walk and ask the children to name as many of these kinds of animals as they can. Put up the pictures of the animals they name, and then add the others. Group the pictures of the mammals separately from those of the reptiles.
4. Ask if anyone would like to choose a picture and talk about the animal in the picture. Ask questions to help the child and invite other children to add information. Also encourage the group to ask questions, and add information when appropriate.
5. After each child has spoken, ask questions to help the children compare the different animals.



6. *Things to talk about each animal:*

- *What it is called (its name).*
- *What it looks like (appearance) and the different parts of its body; what the body is covered with; what the young animals look like.*
- *How it moves and whether it moves fast or slowly.*
- *The sounds it makes; high or low, loud or soft.*
- *How dangerous it is; how it can hurt people.*
- *What it eats and when (some animals only eat at night).*
- *Where it lives or how it is protected; whether it is warm-blooded or cold-blooded.*
- *How babies are produced (live birth or eggs) and how they are cared for.*

7. *Additional Questions*

- *Where have you seen one like this before? How did it look? How does it move?*
- *How do you think its body feels?*
- *Why do you think it has...(e.g. such a long tongue / no legs / a shell on its back / quills)? What do you think it eats?*
- *Which of these animals live in holes in the ground?*
- *Where do the others live?*
- *In what ways are these animals the same? (e.g. most of them have 4 legs; small mammals have fur and give birth to their babies; reptiles have scales and lay eggs) Which animals move fast and which move slowly?*



Animal Homes Classification

Objective:

- To compare the different kinds of small animals according to the homes in which they live.
- To develop classification skills.

Materials Needed:

- All the drawings and picture cards of small animals
- Three drawings of “homes” (sky, ground, water)
- Flannel board, easel, blackboard or wall

Preparation:

1. Color in and cut out the drawings of “homes” and paste onto card.

Instructions:

All Ages:

- Put the 3 “homes” cards on the board and spread the animal cards out in front of the children.
- Talk about where animals live and what they need to live in water, air or land. Ask each child to choose an animal picture and put it under the card showing where it lives.
- As each picture goes up, discuss the animal and the way its body is adapted to live in its surroundings, e.g. birds have wings to fly with and feet to perch on branches or swim in water.
- Ask some or all of the following questions:
 - *What kind of body covering do you think it needs to live in the trees and in the sky?*
 - *Why do you think the lizard has short legs? What would happen if it had long legs?*
- *Can the fish live on land? Why not?*
- *Why do you think moles live underground?*



Discovering About Birds

Objectives:

- To increase children`s understanding of birds..

Materials needed:

- Variety of feathers from different kinds of birds
- Old nest(s), eggshells, seeds and other things they eat,
- Bowl of water
- Drawings and pictures of different kinds of birds and things they eat (insects, worms, snails, mice).

Instructions:

All Ages:

1. Take the children outside to observe the birds quietly for a few minutes.
2. When you return let the children handle the nest, feathers and eggshells.
3. Talk about the feathers and discuss why they are different colors (sometimes to protect them from their enemies) and how they feel. Put the feathers in the bowl of water. See if the feathers will float. Discuss why they float and what feathers are for.
4. Talk about the ways in which all birds are the same: how birds fly, their nests, how the eggs are kept warm and the adult birds care for the young.
 - Then talk about the differences between birds in color, size, how they make nests, how they get food and the kinds of food they eat; differences in the shape of their beaks and in their feet.
 - Ask the following questions:
 - *How do you think birds fly? Can we fly? Why not?*
 - *What would happen if a bird did not have wings? Why?*
 - *What would happen if birds had no feathers?*
 - *Where do you think the birds are flying to?*
 - *What do you think the nest is made of? How?*
 - *Why do you think ducks have webbed feet?*



Animal Movement and Sounds

Objectives:

- To compare the ways different animals move and the different sounds they make..

Materials needed:

- None

Instructions:

INTERMEDIATE/ADVANCED:

1. Combine various movement activities with talking about the different ways animals move and the sounds they make.
 - *Movement:*
 - a. Begin with a movement activity involving all the children in imitating animal movements,
 - e.g. *Waddle like a?* (duck)
 - *Slither on your tummies like a* ? (snake)
 - *Swim like a ...?* (fish) *Now like a tadpole.*
 - *Walk very slowly on four legs like a.....?* (tortoise)
 - *Jump like a....?* (frog)
 - *Fly like a?* (bird)
 - b. Stand in a circle and play a guessing game. Ask individual children to imitate the way a particular small animal moves. The rest of the group must guess what animal it is.
2. Discuss what helps different animals move in certain ways. Talk about which animals move in the same way and which move differently from each other.



- *Why do you think the snake has no legs? How does it move?*
- *What does a hare's feet look like?*
- *How does a fish swim? Can we swim?*
- *Which animal jumps the highest? Why?*
- *What would happen if a lizard had long legs?*
- *Does a lizard move in the same way as a tortoise? In what ways are they the same?*
- *In what ways different? Which animals move fast? Which ones are slow?*

3. Sounds

- Play the Animal sounds game. Discuss the different sounds that animals make, how they make them and the reasons for the different sounds.
- Suggest that the children have animal conversations. Ask other children to guess what they are saying.
- *Ask some of the following questions:*
 - *Do you think all animals make sounds?*
 - *Which animal do you think makes the loudest sound?*
 - *Which animal makes the softest sound?*
 - *What do you think they are saying to each other?*
 - *What do you think would happen if animals could not make any sounds?*

B. IDEAS FOR THE SMALL GROUP ACTIVITIES



Group Nature Collage

Objectives:

- To encourage children to work together and learn to share.
- To develop children's creativity through using natural materials.

Materials needed:

- Newspaper and/or large pieces of the paper
- Leaves
- Wood
- Dry grass
- Seeds
- Crayons or paint and brushes
- Scissors
- Natural materials such as twigs
- Bark
- Soil
- Feathers
- Smaller pieces of paper or card
- Paste/Glue

Preparation:

1. Stick large pieces of paper (on newspaper) together to make a large long piece. Fix the large piece to the wall at the children's height or lay it out on newspapers on tables, the floor or on the verandah. If paper is not available, children can make the collage on the ground.
2. Stick a few items onto the paper that will stimulate the children's imaginations, e.g. piece of blue paper for a dam, some leaves and bark for a tree.

**Instructions:****All Ages:**

1. Encourage children to talk about the plants and animals they have seen in field or near the river. Suggest they use the materials provided to make a group picture. They can also draw or paint, cut and stick things on.
2. Allow the children to work in their own ways and use their own ideas. Some children may not try to make a picture of what they have seen.
3. Encourage children to talk to one another as they work as this will encourage cooperation and the sharing of ideas.

**String Painting****Objectives:**

- To experiment with a different way of putting paint on paper and making “snake” pattern.
- To practice small muscle control.

Materials needed:

- Flat containers, such as styrofoam food trays
- Paint of different colors (if available) or mud paint
- Pieces of string of different lengths and thickness
- Paper

Instructions:**All Ages:**

1. Pour the paint into the containers. The activity is likely to be messy, so lay newspapers down where the children will work and have a wet cloth or water nearby.
2. Explain to the children that they dip a piece of string into the paint and then pull drop or roll it across the paper to make patterns.
3. Encourage them to try different ways of using the string to put the paint on the paper and make patterns.



Observing Small Animals

We do NOT encourage children to catch live animals or keep small animals in cages or other containers. But there are some small animals that can be kept for a few days in the playroom or playground/garden if we can provide a suitable home for them.

Objectives:

- To provide opportunities for children to observe small animals closely and to discuss the results of their observations.
- To stimulate children's interest in small animals and nature.

Materials needed:

- A suitable environment in which to keep the small! animal(s). This must include the following:
 - enough space in which to move,
 - natural materials that are part of the usual surroundings in which the animal lives, such as soil, small stones, grass, leaves and other plants for food,
 - shade and shelter, if necessary,
 - air to breathe,
 - enough food and water.

Instructions:

All Ages:

1. Tortoises can be kept in a safe place in the playground if there is enough grass for them to eat and some shade from plants or trees. If the playground is not enclosed by a fence, the tortoise may escape. Make sure that any enclosure is large enough and that the tortoise(s) have enough food and water at all times.
2. A chameleon can be brought to the playground on a stick and placed on a suitable plant where the children can observe it. Do not try to keep it enclosed in anything.
3. Fish can be kept in a proper fish tank. The fish, tank and food will probably have to be bought at a pet shop. They need the right kind of food, and the water has to be partly changed regularly (if it is not aerated). Also ask for information on the care of fish at the pet shop.



Observing Small Animals: Tadpoles

Objectives:

- To increase observational skills.
- To increase small muscle coordination.

Materials Needed:

- Net(s) for catching tadpoles and a large glass jar (make a net by fixing a piece of orange bag netting onto a wire circle with quite a long handle)
- Stream, furrow or pond nearby where there are frogs and tadpoles

Instructions:

All Ages:

1. Take a small group of children with you to look for tadpoles in a nearby stream, furrow or pond.
2. Prepare the large jar first. Put some sand and small stones from the pond or stream in the bottom of the jar, add water and find a water weed or two to put in the jar.
3. Catch only two or three tadpoles in the net and put them into the jar straight away.
4. Place the jar in a cool place where the children will be able to look at the tadpoles swimming around.
5. Remember that tadpoles need air to breathe, so do not leave a lid on the jar. Feed them every few days on grains of hard-boiled egg yolk.
6. As soon as the front legs begin to appear on a tadpole, it needs to be able to get out of the water, so a jar is not very suitable. Either take the tadpoles back to the pond or stream or transfer them into a larger, flatter container with some stones or small logs of wood in it that rise above the level of water.
7. Do NOT keep frogs in a jar or container. If the tadpoles live long enough to turn into frogs, take the frogs back to the stream or pond.



Spot the Small Animal

Objectives:

- To encourage children to identify similarities and differences and to observe the details.
- To help children develop the perceptual skills of differentiating something from its background.

Materials needed:

- A large piece of thick cardboard from a large carton about 30 x 42 cm
- Poster drawing of woodland scene
- Separate drawings of the small animals in poster
- Small pieces of thin cardboard for the animal drawings
- Crayons,
- Paints
- Scissors
- Paste/Glue
- Plastic contact or varnish (if available)
- Storage box

Preparation:

1. Color in the drawing of the woodland scene and stick it onto the large piece of cardboard. Use realistic colors.
2. Cut out the animal drawings and paste them onto small pieces of cardboard.
3. If possible cover all the pieces and the game board with clear plastic contact or paint with clear varnish to strengthen and protect it.

BEGINNER:

1. Have the children identify specific animals in the big picture.

INTERMEDIATE/ADVANCED:

1. Tell the children to place a similar animal drawing on each of the animals on the game board until all the animals in the picture are covered.
2. If a group of children are playing together they should take turns to find an animal in the picture and cover it with a drawing of the same animal



Animal Homes

Objective:

- To learn the difference between different animal habitats.
- To help children identify animals and their homes.

Materials Needed:

- A large piece of thick cardboard (e.g. from a box)
- Drawing of 5 animals and 5 'homes'
- Crayons or paint, scissors, paste
- Wool or string, split pins, paper or hair clips or buttons

Preparation:

1. Color in and cut out the 10 drawings.
2. On one half lengthwise of the piece of cardboard, paste the drawings of animals.
3. On the other half of the board, paste the drawings which illustrate places where the animals live (see diagram).
4. Use a split pin, paper clip, hair clip or sew a button onto the board next to each picture.
5. Cut 5 pieces of wool or string about the length of the cardboard. Tie one end of each piece of wool/string around the split pin head or button next to each animal picture.

Instructions:

BEGINNER: INTERMEDIATE: ADVANCED:

1. The child fastens the loose end of each piece of wool/ string next to the picture showing where the animal lives.
2. Make sure that the children understand what each drawing is of, especially the drawings of the homes. Help children find the right home by asking questions e.g.
 - *What is this animal? (pointing to the snake)*
 - *Do you know where the snake lives? Does it live in the grass or in the ground? That's right, in a hole in the ground. Now can you find the picture that looks like a hole in the ground?*



Discovery Table

Objectives:

- To encourage children to explore and learn more about their surroundings.
- To develop language skills by encouraging children to share information.

Materials needed:

- Table or large flat box, smaller boxes
- Magnifying glasses, if available

Instructions:

BEGINNER:

1. Ask children to bring things related to small animals that they find in their surroundings and put them on a discovery table, e.g. things that animals eat such as nuts and seeds, grass; feathers, an old nest or egg shells; a shed skin from a reptile, tortoise shell.

***NOTE: It is important that children know that they must not disturb or harm small animals, e.g. they must not take a nest from a tree or eggs from a nest. They must not bring live animals for the table.*

2. Encourage children to talk about what they have brought, where they found it, what animals eat it, what they know about the animals and their habits

INTERMEDIATE: ADVANCED:

1. Ask children to bring things related to small animals that they find in their surroundings and put them on a discovery table, e.g. things that animals eat such as nuts and seeds, grass; feathers, an old nest or egg shells; a shed skin from a reptile, tortoise shell.

***NOTE: It is important that children know that they must not disturb or harm small animals, e.g. they must not take a nest from a tree or eggs from a nest. They must not bring live animals for the table.*

2. Encourage children to talk about what they have brought, where they found it, what animals eat it, what they know about the animals and their habits.
3. Children can also sort (classify) the different things into smaller boxes, e.g. all things to do with birds in one box; or different kinds of animal food in each box.



C. IDEAS FOR OUTDOOR ACTIVITIES

1. Take a walk in the neighborhood and look for community helpers. If possible have the children talk to them, asking them what they do, how they like their job, what their training was for the job, etc.
2. Take a walk in the neighborhood with the intention of finding a particular community helper. Focus on who you will be looking for at Morning Meeting. Have the children search for the person on their walk. When they find the helper, have the children watch what he or she is doing. Go back and “draw” what they have seen. For older children, have them dictate a story about their walk. Put all of the children’s drawings together to make a book that can be put in the Literacy Learning Center for “reading.”



Nature Walk

Objectives:

- To appreciate their environment.
- To practice visual and auditory discrimination.
- To stimulate oral expression.
- To practice new vocabulary.

Materials needed:

- None

Preparation:

1. During morning ring or group time the day before tell the children that they will be going for a nature walk. Ask them what kinds of small wild animals live in the veld.
2. Explain that they might not see any small wild animals, except perhaps birds, on the walk, but that there are signs that they are close by. Discuss the sorts of things that they will be looking and listening for, e.g.
 - listen for sounds of animal noises
 - look for different homes of animals, e.g. mole-hill, tunnels, holes, nests, dams
 - food that animals eat, e.g. seeds, worms
 - signs that animals are around, e.g. tracks, feathers, porcupine quills
 - birds and how they fly.

Instructions:

During the walk, encourage the children to walk quietly, look carefully and listen for animal sounds.

- *All stop and listen very carefully.*
 - *What noises can you hear?*
 - *What kind of animal makes these footprints/tracks?*
 - *What do you think lives in this hole? What color is that bird?*
2. Pick up feathers, quills, and so on to take back to the playroom.
 3. **Talk-about:** After you return, in group time, discuss the things you have seen and heard. Children could tell a story, e.g. "Today I went for a walk and I saw / heard..." Each child completes the sentence.
 4. Use pictures of the animals that the children saw, heard, found tracks of or any other evidence. Put the pictures up on the wall or flannel board, and place things like feathers or quills next to the relevant picture.
 5. Encourage the children to talk about the different small animals that were identified during the walk and about other small animals that they know about. This will give the teacher an idea of how much the children know.
 - *What did it look like?*
 - *What sound did it make?*
 - *What do you think lives there?*
 - *How do you think it is made?*
 - *Why do you think it....? (e.g. lives in a hole in the ground)*

Note: With **younger children** talk about only the animals that children have actually seen.

**Extension:**

1. Take children on walks to different places where they may see specific small animals, e.g. a river or dam (frogs, tadpoles, and fish), the sheds (mice, bats) or a place with many trees or bushes.
2. Take children on walks to look for specific animals, e.g. a bird walk, or a walk to look for animal tracks (footprints) or animal homes (holes, nests, etc.).
3. Link these walks with special activities, such as looking for a tortoise (page 7-8), and/or with talk-about times in the following pages.

**Catch the Snake's Tail****Objectives:**

- To improve large muscle coordination.
- To stimulate imagination.

Materials needed:

- None

Instructions:

1. Children form a chain by each one holding the waist of the child in front of him/her. The child in the very front is the "head" of the snake, and the child at the back is the "tail". The aim of the game is for the head to catch the tail., but the tail tries not to be caught. When the teacher says "go", the head chases the tail. Once the tail has been caught, the child at the head can join the tail end, and the next child becomes a head.



TOPIC V: ANIMALS: TINY CREATURES

BACKGROUND INFORMATION

The topic “Tiny Creatures” is about animals which do not have backbones. It focuses on insects and other tiny creatures such as spiders, snails and worms that live around us – in the ground, under stones, in the grass, near water, on plants and trees, in the air and even inside buildings. It extends children’s knowledge of animals but these are very different from farm animals and small wild animals which all have backbones, even snakes.

One of the best ways for children to learn more about these tiny animals is through observation. Take children on walks in their immediate surroundings. Almost everywhere you look you will find some sign of insects and other tiny creatures. Look under plant pots, in cracks in the paving, on the walls, in the grass, on leaves, under stones.

Children should be encouraged to observe them in their natural surroundings most of the time, but some of these creatures can be kept in the playroom in suitable containers and observed for a short period before returning them to their natural environment. However, if you choose to keep them for longer periods so that you can find out more about them, you will have to know something about their natural environment to be able to care for them properly and know how to feed them.

It is important for teachers to show children by example how to observe and respect nature. For example, do not touch, hurt or kill insects unnecessarily; show children how to collect insects for observation without handling them (e.g. by using suitable containers, use a spade to lift ants with soil, break the twig or leaf off with the caterpillar on it). Children should also learn which ones can hurt them, e.g. bees and wasps sting; scorpions and centipedes are poisonous; flies spread disease.

The following information is provided for teachers to use when children are interested and able to understand it through their own observations. Teachers’ aim is to arouse children’s curiosity and to help them find things out for themselves. They should not have to memorize many facts!

Insects – e.g. *flying insects such as bees, butterflies, moths, flies, dragonflies, ladybirds, and crawling insects like ants, beetles, crickets, caterpillars.*

- All insects have bodies that are divided into three parts. The front part is the head which has eyes and *antennae (feelers)*. The middle is called the *thorax* – the legs and wings are attached to this. They breathe through two holes in the thorax called *spiracles*. The rest of the body is called the *abdomen*.



- All insects have 6 jointed legs (3 pairs) and if they fly, they have 2 pairs of wings (4). If you are not sure whether something is an insect, count the legs.
- A baby insect is often not at all like the adult. It may have to pass through several changes in its life-cycle. Insects lay eggs which hatch into *larvae* (e.g. caterpillars, maggots, grubs). This change into pupae and the adults finally hatch out of the pupae. This complete change is called *metamorphosis*.
- Insects outgrow their body coverings which are usually hard (external skeletons to hold the body together). They shed the one that is too small and grow a new, larger one.
- Many insects use their legs for more than walking – they use them for swimming, jumping or digging. Some have forelegs that are adapted for grasping and holding prey (e.g. the praying mantis). Others have legs with hairs or suction pads to help them walk on water.
- Insects do not have a voice, but they do make sounds with other parts of their bodies. Some may hum or buzz by vibrating their wings, others may squeak by forcing out air through their breathing holes. Or they may make a sound by rubbing one part of their body against another part.
- Many insects only eat seeds and plants, but some eat other insects or suck blood from humans and animals.
- Insects provide food for people (honey, locusts) and other animals, and help to spread seeds and pollen (plants). Some insects are pests (eat crops) or spread diseases.

Spiders and scorpions

- Spiders and scorpions belong to the *arachnid* family which also includes ticks and mites. All arachnids have 8 legs (4 pairs) and no antennae (feelers).
- Most spiders spin silk to make webs for their homes or to help catch insects for food; they wrap their eggs in a silk cocoon.
- Spiders help us by eating insect pests such as flies. Sometimes they chase the insects they will eat and spit sticky material onto their victims. Other spiders wait in holes for insects to fall in.
- Spiders only attack (bite) people if they are provoked or in danger. A few spiders are very poisonous.
- Scorpions eat insects; they live under stones; they have a pair of pincers in front and a poisonous sting in their long tails.
- Ticks and mites are parasites that live mainly on animals. Tick bites can make people ill (tick bite fever).

Worms (e.g., centipedes, millipedes, earthworms)

- Centipedes are flat worms with many legs; they live under stones; they eat tiny creatures, and some are very poisonous.
- Millipedes (e.g. songololo, but some kinds are much smaller) have round, hard bodies with hundreds of legs; eat plants.
- Earthworms live in the soil and their tunnels help to drain and aerate the soil. This provides good conditions for plant roots. They feed on decaying food matter, and are eaten by birds.

Snails and slugs (*mollusks*)

- Snails have a shell to protect themselves, and slugs live underneath stones and in thick damp vegetation. They also have feelers and eyes on the end of stalks. A snail produces slime which makes its movement easier.

WEEKLY PLANNING OVERVIEW – June, Week 4

V: Tiny Creatures

Time	Type of Activity	Monday	Tuesday	Wednesday	Thursday	Friday
9.00 – 9.15	Morning Meeting	• Welcome children • Greetings • What is the topic this week? • Crawling Insects p. 81	• Welcome children • Greetings • What is the topic this week? • Flying Insects p. 82	• Welcome children • Greetings • What is the topic this week? • Spiders & Insects p. 83	• Welcome children • Greetings • What is the topic this week? • Snails & Slugs p. 85	• Welcome children • Greetings • What is the topic this week? • Worms p. 86
9.15 – 11.00	Work Time – Learning Centers: Teacher organizes materials for children to work in centers of the child's choice including: Art Center, Block Center, Drama Center, Literacy Center, Manipulative Center while one or two Small Groups are working on teacher-initiated activities.					
	Small Groups Week 4	Insect Collection p. 89	Rock Painting p. 91	Spider Web Collage p. 92	Rock Painting p. 91	Spider Web Collage p. 92
11.00 – 11.15	Clean-Up					
11.15 – 12.00	Outdoor Time	Insect Hunt p. 100	Insect Keeper p. 96	Insect Hunt p. 100	Insect Keeper p. 96	Spider Web p. 105
12.00 – 12.45	Wash-Up, Lunch					
12.00 – 12.45	Rest & Quiet Time	Children who can't sleep may "read" books.				
14.15 – 14.30	Group Story Time	• Social games, story reading, finger plays, singing, dramatization of the books, reading the children's books related to the topic. • Discovery Box: exploring sounds, tastes, smells, things found in nature and manufactured • Movement game "Touch blue..." • Asking questions about what did children find during Sense Walk				
14.30 – 15.00	Closing Meeting	• Review of the day, sharing impressions of the day, discussion of the activities children have been involved in – which new things they learned and planning the next day.				
Activities with Parents throughout the day		• Parents can assist in visiting the classroom and telling about their job/role • Parents can assist children in making their postcards • Parents can assist as volunteers in a field trip				



A. IDEAS FOR THE MORNING MEETING



Crawling Insects

Objective:

- To explore the similarities and differences between crawling insects by observing and discussing their behavior.

Materials needed:

- None

Instructions:

BEGINNER: INTERMEDIATE: ADVANCED:

1. Begin with related song, rhyme or movement activity. Perhaps they can crawl like ants.
2. Provide different kinds of crawling insects in jar and pictures children to look and compare the similarities and differences. Note that they all have six legs and two feelers. Some have wings but they do not often use them. Many have hard body coverings, though they may look quite different. Talk about what they eat.
3. Ask the following questions to get the children to think about what insects do:
4. What do you think the ants are doing?
5. What do ants/stink bugs eat?
6. What would happen if they did not have feelers?



Flying Insects

Objective:

- To extend children`s knowledge of the life cycle of insects and compare various flying insects.

Materials needed:

- Dead or alive (in suitable containers) flying insects and things that are associated with them, e.g., empty cocoons, honeycomb
- Pictures
- Pictures of the life cycle of the butterfly

Instructions:

BEGINNER:

1. Have children look at pictures of insects and talk about the name of the insects. Discuss colors, wings, legs, etc., other features they may notice.

INTERMEDIATE: ADVANCED:

1. Talk about the life cycle of the butterfly using the pictures. Encourage children to ask questions and share what they have observed about caterpillars and butterflies.
2. Compare insects like butterflies and moths with other kinds of flying insects. Identify the similarities, e.g. they all have 4 wings, and talk about some of the differences.
3. The children might be interested in learning more about bees which are also social insects like ants. Bees provide honey which is good for people.
4. Children can imitate the sounds that different insects make, such as bees, crickets, cicadas, grasshoppers.
5. Ask the children questions about insects.
 - *What is inside the butterfly`s eggs?*



- *Why do you think the caterpillar has so many different colors? What does it eat?*
- *How does the caterpillar change into a butterfly?*
- *What do you think the butterfly eats?*
- *In what ways are flying insects the same?*
- *Why don't people like flies?*
- *Why and how do people farm bees? Where do wild bees make their homes (honeycombs)?*
- *What do grasshoppers eat?*
- *What sounds different insects make?*



Spiders and Insects

Objective:

- To compare spiders, scorpions and ticks with each other and with other insects.

Materials needed:

- Dead or alive specimens (in suitable containers)
- Pictures

Instructions:

All Ages:

1. Divide the class in half so that one group moves as the spiders, slowly stretching into wide spiders' web shapes on one spot. The other children are the flies; they run in and out of the spider's webs as music plays. When the music stops, a "fly" is caught in a web if the child is between the stretched spider shapes.

2. After the game, talk about how the spider makes a trap with its web for flies and other insects. Talk about where you would find these webs; whether children can see any in the playroom, and what happens to the fly once it is caught. Tell the children about how the spider wraps its eggs in a cocoon made of silk (like a web). Note that baby spiders hatch out of the eggs (no metamorphosis like insects).
3. Compare different kinds of spiders that children are familiar with. In what ways are they the same and different? Children may not be familiar with scorpions, but they need to know what they look like because they can harm them. They may have seen ticks on animals, and know that they suck blood (parasites).
4. Discuss the following questions with the class:
 - *How does the spider make its web?*
 - *What do you think it feels like?*
 - *How do spiders help us? (catch flies)*
 - *How can scorpions hurt people, animals and insects?*
 - *What do you think it eats?*
 - *How can you tell that a creature is a spider/scorpion?*
 - *In what ways are they different from insects (8 legs)*
 - *Do they have feelers?*



Snails and Slugs

Objective:

- To learn more about different kinds of animals.

Materials needed:

- Snails and slugs from the garden, empty shells
- Sheet of glass (or use a window, or glass jar)
- Some green leaves
- Warm water in a container

Instructions:

All Ages:

1. Children could help to find snails and slugs by going on a quick hunt in nearby garden. They like damp places with thick vegetation. Take suitable containers (a plastic bag will do).
2. Place the snails and slugs on the piece of glass (or in a jar) so that children can compare them. They can look at them from underneath. They will be able to see the snail's mouth. Talk about what they look like and how they move.
3. Put the leaves out and ask children what they see and hear – if they listen very carefully they will be able to hear it scraping away with its rough tongue. (If the snail does not want to come out of its shell, dip it quickly into warm water and it will gradually appear).
4. If you have more than one snail you could have a “snail race”. Put out a lettuce or other green leaf and see which snail gets to it first.

5. After the group time, two children can take the animals back to where they found them.
6. Ask/discuss the following questions about snails and slugs:
 - *What is different about snails and slugs from any other creature?*
 - *What do you think the two pairs of horns are for?*
 - *How do they move? Do they move in the same way?*
 - *Why do you think they have no legs?*
 - *What would happen if the snail did not have a shell on its back? How do you think the slug protects itself?*



Worms

Objective:

- To compare different worm-like creatures with each other and with other animals.

Materials needed:

- Earthworms in a box of soil
- Dead and alive specimen of other kinds of worms in suitable containers, such as millipedes and larvae (e.g. caterpillars, grubs, fly maggot, worms in fruits)
- Avoid live centipedes (fewer legs) – very poisonous
- Pictures

**Preparation:**

1. Set up the earthworm box two days before so that they have had time to settle down and make tunnels.

Instructions:**BEGINNER:**

1. Begin with asking children to move like worms. Draw attention to those who are imitating a particular kind of worm, e.g. an earthworm moves very differently from a caterpillar. Can they curl into a ball? Ask the children why they move differently.
2. Perhaps the children can then worm their way to the earthworm house to have a look at what they do. Talk about what they look and feel like and how they help things grow by aerating the soil.

INTERMEDIATE/ADVANCED:

1. Begin with asking children to move like worms. Draw attention to those who are imitating a particular kind of worm, e.g. an earthworm moves very differently from a caterpillar. Can they curl into a ball? Ask the children why they move differently.
2. Perhaps the children can then worm their way to the earthworm house to have a look at what they do. Talk about what they look and feel like and how they help things grow by aerating the soil.
3. Talk about the various kinds of worm specimens and pictures. Compare their similarities and differences and talk about what they eat, where they live, and so on. Note that only caterpillars change into something else. Explain what a centipede is like (flat, many legs, lives under the stones) and that it is dangerous.
4. Ask these questions and any others you can think of about caterpillars and earthworms:
 - *Why do you think that caterpillars roll themselves into a ball? Do caterpillars have legs?*
 - *In what ways are earthworms the same/different from caterpillars and other animals?*



Insect Guessing Game

Objectives:

- To use their memory.
- To learn guessing techniques and putting together ideas.
- To reinforce prior learning about insects.

Materials needed:

- None

Instructions:

All Ages:

1. Tell the children to listen very carefully to what you say and try to guess what tiny creature you are talking about.
E.G.
I have four wings and I fly.
You can find me near water.
I drink your blood! Who am I?

I have hundreds of legs.
I am black and shiny.
I curl into a ball when I am frightened.

I like eating flies.
I have 8 legs.
I wrap my eggs in a silk cocoon.
2. Do not use more than three ideas at a time.
3. Encourage the older children to take turns in giving clues for others to guess. They can also imitate movements or sounds as part of your clues.



B. IDEAS FOR THE SMALL GROUP ACTIVITIES



Insect Collection

Objective:

- To encourage children to explore and learn more about their natural environment.

Materials needed:

- Magnifying glasses (if available)
- Boxes for sorting the items into

Instructions:

All Ages:

1. Ask children to bring things they have found in their surroundings that are related to insects and other tiny creatures. These might be things that insects eat old skins or dead insects and other creatures. It is important that they understand that they must not harm or disturb living insects or their eggs.
2. Put the items on a special display table initially so that the children can examine them more easily. If available, have a magnifying glass or two on the table.
3. Add pictures and books to the table which show pictures relating to what children have brought.
4. Encourage the children to talk about what they have brought and where they found it.



Dress-up Wings

Objective:

- To encourage children to use their creativity and memory skills in painting patterns that they have seen on butterflies.

Materials needed:

- Large sheets of paper
- Strips of light cardboard paste or stapler
- Paint and brushes
- Scissors

Preparation:

1. Fold a large sheet of paper in half.
2. Measure the length of a child's arm. Draw a wing shape to fit it, away from the fold.
3. Cut out the wing shapes through both sheets of paper.
4. Cut two narrow strips of cardboard and stick these onto the underside of each wing for the child's arms to slot through (or cut longer strips of card/paper and slits in the wings so that each strip can make a loop and the ends stapled together). It may be necessary to make this section of the wings stronger with cardboard.

Instructions:

All Ages:

1. Children paint their wings and put them out to dry.
2. When the wings have dried, children can wear their wings and "fly" during outdoor play time.



Rock Painting

Objective:

- To create colorful creatures from the stones that children can find outside.

Materials needed:

- | | | |
|-----------|---------------|----------------|
| • Stones | • Twigs | • Leaves |
| • Sand | • Pebbles | • Paint |
| • Brushes | • Strong Glue | • Masking Tape |
| • Bags | | |

Preparation:

1. Take the children on a rock hunt in the neighborhood. Ask them to look for interesting shaped stones (not too big – about the size of a child's fist)
2. Put some masking tape on each stone and label with the child's name. Collect the stones in strong bags.
3. The children wash the dirt off the rocks and dry them.

Instructions:

BEGINNER: • Note: Too difficult for younger children

INTERMEDIATE/ADVANCED:

1. Set out the paint, glue and other materials with the stones that the children have collected. As each child comes to the activity, she or he collects his or her rock.
2. Talk to the children about insects that they might have seen during their outing and encourage them to make their own little creatures from the materials. Children should be free to create in their own way, but the teacher can ask questions that will help the children, e.g. *How many wings does your butterfly need?*



Spider Web Collage

Objectives:

- To explore making web patterns in a collage
- To develop planning skills.

Materials needed:

- Paper, pieces of wool or string paste, scissors pencils, ruler, rubber

Instructions:

BEGINNER:

1. Talk to the children about what a spider web looks like. Ask them if they could make a collage with the pieces of wool/string to look like a spider web.
2. Do not show them how to do it and allow them to make other patterns if they wish.

INTERMEDIATE/ADVANCED:

1. Talk to the children about what a spider web looks like. Ask them if they could make a collage with the pieces of wool/string to look like a spider web.
2. Encourage the children to plan in advance how they are going to make the web pattern. They can either lay the pieces of wool out before putting paste on or draw the web with a pencil. *What would be the easiest way to put the paste on? On the wool or on the paper?*
3. Do not show them how to do it and allow them to make other patterns if they wish.



Egg Box Caterpillars

Objective:

- To practice fine-motor skills to make and decorate a caterpillar.

Materials needed:

- | | | |
|----------------------------|------------------------------|----------------------|
| • Cardboard egg boxes | • Twigs | • Wool |
| • Colored paper | • Material scraps | • Buttons |
| • Paste/glue | • Hole punch or a large nail | • Scissors |
| • String or wool or sticks | • Sticky tape | • Markers or crayons |

Preparation:

1. Cut the lower half of the egg boxes into individual sections (cups).

Instructions:

BEGINNER:

1. Put out materials and provide children with various art materials. Give each child a piece of paper and tell them to draw /construct a caterpillar.

INTERMEDIATE/ADVANCED:

1. Set out the egg carton sections with the other materials.
2. Ask the children if they can think of a way to make caterpillars from the materials provided. Encourage them to talk about different ways of doing it. They can either link the sections with wool/string or sticks.
3. Children select their materials and decorate them as they want.
4. Help the children make holes in both sides of the sections (except for the head), either with a punch or nail. Threading wool or string through the holes will be easier if the end is hardened with sticky tape (or glue).



Observing Tiny Creatures

The best way to observe insects and other tiny animals is to watch them quietly in their natural surroundings, but some kinds can be collected and kept in suitable containers for a few hours or days and sometimes longer.

Objectives:

- To provide opportunities for children to observe tiny creatures closely and talk about them.
- To stimulate children's interest in nature.

Materials needed:

- Suitable containers in which to keep each type of tiny animal, which must provide the following:
 - enough space in which to move,
 - natural materials that are part of the usual surroundings in which the animal lives, such as soil, stones or twigs
 - shade and shelter, if necessary
 - air to breathe
 - enough food and sometimes water.

Instructions:

All Ages:

1. Always take great care when collecting tiny creatures. Try not to handle them directly, e.g. collect ants using a trowel to pick up some soil as well; cut the twig with the leaves the caterpillar(s) are on; cover a flying insect or spider on the wall with the jar and slide a piece of paper underneath it. Show the older children how to collect some of insects and other tiny creatures.
2. Do NOT collect poisonous or dangerous creatures, like poisonous spiders, thick-tailed scorpions (small pincers) or centipedes. Be very careful with stinging animals like bees, wasps or thin-tailed scorpions (big pincers).



3. Do not put the containers in the full sun, especially if there are no leaves in it, although many tiny animals like a warm place. Animals that live in the soil need damp leaves to keep them damp (or they will soon die).
4. Do not frighten the animals by shaking or banging the containers.
5. If you have a magnifying glass, the children can use it to examine the creatures. Show children how to hold the glass in front of the eye - make sure that they do not hold the glass at arm's length.
6. Always return the small creatures to the place where they were found.

1. Few hours:

- Many kinds of insects, such as beetles, caterpillars, ants, flies, bees, snails, spiders and millipedes can be collected in suitable jars and kept for a few hours for children to observe.
- A plastic or glass jar will be adequate if it is big enough. Punch some holes in the lid of the jar to allow air to enter the jar (or tie a piece of panty hose over the open top - use for ants).
- Include some of the leaves of the plant which the animal was eating when found.
- Do not mix different kinds of creatures in one jar unless they were on the same plant.

2. Few days:

- It is very important that suitable homes and care are provided if tiny creatures are kept for a few days or more. Children need to be aware of the shelter and food required by the creatures so that they may be unharmed and returned to their environment. Provide water by putting a wet sponge in the jar (needed by animals which do not eat plants).
- Snails can be kept in a jar covered with netting (orange bag), panty hose or a lid with holes in it. Give it fresh leaves (preferably from the plant where you found it) twice a day (snails feed mostly at night, so make sure there is enough) and clean out the jar every day or two. Keep the jar in a cool place. Look at the two pairs of horns on the snail's face. The two long ones have eyes on the end; the other two are for feeling.

3. Observing Tiny Creatures for a Few Days

- Earthworms are very easy to find - you can dig them up with the soil. Keep them in a strong box, preferably wood because the soil must be kept damp. Cover it with sheet of plastic so that the children can observe how the worms burrow (like the Ant Farm idea on page 8). Fill the box with damp soil - this will also provide food. Keep the soil damp and do not keep the worms for longer than a few days.

*Ants, caterpillars and silkworms can be kept for longer periods. Ideas are suggested in the following pages.



Insect Keeper

Objective:

- To keep tiny creatures for observation.

Materials needed:

- Large plastic bottle (e.g. 2 liter coke bottle)
- Old stockings or panty hose
- 2 rubber bands or string, scissors
- Collection of insects

Instructions:

All Ages:

1. Cut around the bottle just below the opening.
2. Cut out four “windows” in the sides.
3. Slip the bottle inside the panty hose.
4. Cut off the feet of the panty hose so that both ends are open.
5. Gather the stocking to one side at the bottom and fasten it securely shut with a rubber band or knot.
6. Fasten the top with a rubber band once the insect has been found.



Butterfly Sequence Cards

Objective:

- To develop thinking skills related to sequences.

Materials needed:

- 6 drawings of the life cycle of a butterfly
- Pieces of cardboard {e.g. from soap powder boxes}
- Crayons or paints, scissors and paste storage box

Preparation:

1. Color in the drawings with crayons or paints.
2. Stick the drawings onto the cardboard and cut them into cards. Strengthen the cards with clear varnish if this is available - they will last longer.
3. Store the cards in a suitable box.

Instructions:

BEGINNER: • Note: Too difficult for younger children

INTERMEDIATE AND ADVANCED:

1. The child puts the cards together in sequence working from left to right, starting with the butterfly laying her eggs until the new butterflies fly around.
2. Teachers can help children to understand the life cycle of the butterfly by talking about what each picture represents and asking questions like:

- *What do you think the butterfly is doing in this picture? (laying eggs)*
 - *What comes out of the eggs when they hatch?*
 - *Which picture shows the baby caterpillars hatching?*
 - *What are the caterpillars doing here? (eating a leaf)*
 - *What do you think is happening in this picture?*



Ladybird Card Game

Objectives:

- To practice recognizing number sets up to ten.
- To encourage children to develop their visual perception and memory skills.

Materials needed:

- Pieces of cardboard (e.g. from cereal boxes)
- Ladybird drawings (10)
- Paint or crayons, scissors paste
- Storage box

Preparation:

1. Color in the drawings of the ladybirds (either all red with black heads or different colors that do not match the dots) and stick them onto the pieces of cardboard.
2. Cut out the 10 cards and then cut each through the middle on the dotted lines. Make sure that all the edges are stuck down, and cover with plastic contact or clear varnish (if available).
3. This game can be played individually or by 2 to 4 children as a group.

**Instructions:****All Ages:**

1. First look at the cards together. Check that the children can match all the number sets correctly. Simplify the game at the beginning if necessary by using the first 6 pairs only (12 cards).
2. Mix up the 20 cards and spread them out face down on a table or on the floor. The first player turns face up any two cards. If they match, she takes them and she turns up another two cards. If the cards do not match, the same child turns them face down again in exactly the same places.
3. The next player then has a chance to turn over any two cards. As the players see the different cards being turned up, they can remember where a particular card is and this helps them, to find the other half.
4. They continue until all the pairs have been found.

C. IDEAS FOR OUTDOOR ACTIVITIES



Insect Hunt

Objectives:

- To introduce the topic on tiny creatures and to find out what children already know.
- To develop children's observational skills.

Materials needed:

- Drawings/posters and pictures of different tiny creatures
- Magnifying glasses
- Suitable containers to collect live specimens
- Walk poster

Preparation:

1. Before you go on the walk, put out a trail of food, e.g. jam or old bones. This will attract insects so that children will be able to observe them at work as they find the food and carry it off.

Instructions:

All Ages:

1. Tell the children that they will be taking a walk to look for insects and other tiny creatures. Ask questions about what creatures they may find on the walk and where they might look for them.
2. Talk to children about how easily these insects can get hurt, e.g. butterflies' wings are very fragile and insects' legs can easily be broken. Children should not touch the things they find. They need to approach these creatures very quietly so as not to scare them away. Also talk about creatures that may sting or bite if they are frightened, e.g. a bee or wasp.



3. On the walk look for ants and anthills. Observe the way ants are working and discuss the different jobs they are doing. Look for eggs of insects under leaves (again be very careful that you do not harm any creature). You might find a variety of animals under stones or the bark of trees. Talk about why they need to hide under the bark or under rocks and stones. Look for shed skins of insects.
4. Children can examine ants and caterpillars with a magnifying glass. Ask questions to help them observe more carefully, e.g.
 - *What is it called?*
 - *How many legs/wings does it have?*
 - *How do you think it feels to touch? What color is it?*
 - *What do you think it eats?*
5. Show the children how to carefully collect a few harmless specimens in suitable containers without hurting them to take back for observation and discussion in the playroom.
6. After you return, in group time, encourage the children to talk about what they saw on the walk. Have pictures available so that children can identify the creatures that they saw. Put the pictures up on the wall or on a flannel board. Display the things that were collected and talk about them too.
7. Encourage the children to tell you what they know about the different creatures so that you know what to build on in the following talk-about. They can also talk about insects and other creatures that they have seen elsewhere. Focus on finding out what the children already know, rather than telling them things.
8. They can also look at the walk poster and identify the tiny creatures in it, and the ones they have talked about before (which have backbones).
 - *What sound does it make?*
 - *Why do you think it lives there?*
 - *What would happen if it...(e.g. had no wings)*
 - *Where have you seen one like this before?*
 - *Why do you think...(it has so many legs, eyes)?*



Bees Fly

Objectives:

- To increase categorization skills.
- To use prior learning to reinforce concepts.

Materials needed:

- None

Instructions:

1. The teacher stands facing the children. She calls out bees fly and she flaps her arms as if she is flying. Children do the same because bees do fly. She calls out butterflies fly, moths fly, ladybirds fly and so on, and she and the children flap their arms. When she says the name of a creature that does not fly, e.g. *snails fly*, she still flaps her arms, but the children must not flap their arms. They must stand still.



Tiny Creatures Moving

Objectives:

- To reinforce auditory skills.
- To encourage imagination and creativity.

Materials needed:

- None

Instructions:

All Ages:

2. Adapt a story or tell your own. Each child chooses his or her own movement, and moves freely. The teacher encourages them to think for themselves about how they can dramatize with their actions. Here is an example of a story which you can tell or adapt to suit your own situation:
3. *Listen to the music and do what it tells you. All the creatures are invited to a party. The ants come marching (drum plays a march)... the beetles come running (drum plays quickly)... the snails crawl slowly (drum plays slowly)... the grasshoppers come hopping (drum plays quickly)... How do the spiders walk?(The drum stops) - yes, the spiders stop to look all around them from time to time. The bees come buzzing; flying so fast that you can hardly see their wings. Let's see if you can do that.*



How Insects Change

Objectives:

- To use imagination and creativity.
- To reinforce concepts.

Materials needed:

- None

Instructions:

1. Children dramatize (mime) the story of the life cycle of the butterfly. They begin as eggs (children make themselves into tiny balls). Then the caterpillars hatch out of the eggs and eat and eat until they get too big for their skins (how can the children mime 'shedding their skins'?) The caterpillars continue eating and moving along the plant stalks and leaves (children move like a caterpillar). A bird comes along and they curl up in a ball. When they have eaten enough, they tie themselves to a stalk (stand up with arms around the stalk) and turn into a cocoon (they must stand very still). Finally they come out of the cocoon as a butterfly and fly about.



Spider Web

Objectives:

- To improve large muscle coordination.

Materials needed:

- None

Instructions:

1. Choose one child as the spider and have he/she stand with her back to the rest of the children.
2. Tell the other children stand at the other end of the room. Tell them to start walking, counting their footsteps aloud: 1-2-3-4-5. When they reach 5, they must stand still. The spider turns around quickly, and tries to catch someone moving. Anyone who moves must join the spider in her web. The spider turns her back again and the children count again and move forward. If one of them can touch the spider while her back is turned, then that person becomes the next spider.

